



Washington University in St. Louis

JAMES MCKELVEY SCHOOL OF ENGINEERING



THE
ATLAS

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Acronyms and Nomenclature

α , AoA	Alpha or angle of attack	KM Sim	Kinematic mission simulator
AIAA	American Institute of Aeronautics and Astronautics	K_v	Motor velocity constant
AUW	All up weight (Aircraft weight including payload and passengers)	l_{ac}	Distance from aerodynamic center to the center of gravity
AR	Aspect Ratio	$l_{ac,w}$	Distance from aerodynamic center of the main wing to the center of gravity
CA	Cyanoacrylate	$l_{ac,h}$	Distance from aerodynamic center of the horizontal stabilizer to the center of gravity
CAD	Computer-aided design	LiPo	Lithium polymer
CF	Carbon fiber	M1	Mission 1
CFD	Computational fluid dynamics	M2	Mission 2
CG	Center of gravity	M3	Mission 3
CWL	Cubic wing loading	NACA	National Advisory Committee for Aeronautics
$\bar{c}$	Mean aerodynamic chord	η_h	Dynamic pressure ratio for the horizontal stabilizer
C_D	Coefficient of drag	ρ	Density
C_L	Coefficient of lift	PLA	Polylactic acid
C_{L0}	Coefficient of lift at 0 AoA	PCB	Printed circuit board
$C_{L,\alpha}$	Coefficient of lift with respect to alpha	Re	Reynolds number
$C_{L,\alpha,w}$	Coefficient of lift with respect to alpha on the main wing	RPM	Rotations per minute
$C_{L,\alpha,h}$	Coefficient of lift with respect to alpha on the horizontal stabilizer	S_h	Size of horizontal stabilizer
$C_{l,\beta}$	Coefficient of roll with respect to beta	T	Thrust
$C_{n,\beta}$	Directional stability coefficient	T/W	Thrust-to-weight ratio
$C_{m,\alpha}$	Longitudinal stability coefficient	UBEC	Universal battery eliminator circuit
d	Diameter	V_∞	Freestream velocity
$\frac{dc}{d\alpha}$	Washout factor affecting airflow over the horizontal stabilizer with respect to alpha	ν	Kinematic viscosity
DBF	Design/Build/Fly	$V_{logic, min}$	Minimum signal voltage for smooth servo operation
EPS	Expanded Polystyrene	$V_{logic, abs min}$	Minimum signal voltage for any servo operation
ESC	Electronic speed controller	WU	Washington University in St. Louis
FEA	Finite element analysis	WUDBF	Washington University Design/Build/Fly
GM	Ground Mission		

1. Executive Summary

1.1 Overview

This report describes the design, testing, and manufacturing processes of the *Atlas*: the Washington University in St. Louis Design/Build/Fly (WUDBF) team's entry in the American Institute of Aeronautics and Astronautics (AIAA) 2020 Design-Build-Fly (DBF) competition in Wichita, KS, shown in Fig. 1.1. WUDBF aims to complete all missions while maximizing expected score. To meet this goal, the team uses a three phased design process: conceptual, preliminary, and detail. Conceptual Design shows competition score is independent of aircraft configuration, so a risk-minimizing pylon racer-type design was selected as the team's aircraft. Preliminary Design highlights the dynamic stability of the aircraft, resulting in good handling characteristics as proven in flight tests. It also shows that an aircraft with two motors mounted on the wing balances dynamic and static thrusts. Detail Design describes a wood-based structure with a 5 ft wingspan, electronic integration, and an expected Total Mission Score of 5.82. Primary construction materials are balsa wood, basswood, and UltraCote. Tests show a wood-based aircraft best provides ease of iteration, flexibility, and reliability. Aircraft integration is confirmed in targeted flight tests.



Figure 1.1: The *Atlas*.

1.2 Summary of Design

WUDBF determined that expected score is maximized through balanced optimization of the passenger and banner missions. The *Atlas* is a conventional, high-wing, dual motor aircraft with tricycle landing gear, partially optimized for Missions 2 and 3. Derived from a pylon racer-style aircraft, iterative design changes ultimately resulted in a vehicle resembling a military transport aircraft. The *Atlas* has a high wing and high tail setup, allowing ample ground clearance for propellers, Fowler flaps, external storage, and internal storage via nose cone opening. The propulsion system is designed for high cruise speeds of 35 m/s and 20 m/s during Missions 2 and 3, respectively, while satisfying short takeoff requirements. Designed to carry 21 passengers and luggage, the aircraft uses balsa wood construction techniques. Two flight tests have been completed, confirming flight worthiness and providing preliminary competition performance estimates.

1.3 Aircraft Capabilities

Table 1.1 provides basic information about the *Atlas* design and projected performance based on aircraft CAD models, a kinematic simulator (KM Sim), and flight test data, per competition requirements.

Table 1.1: Basic aircraft information and performance data.

M1 Takeoff Weight [kg]	4.9	Average Cruise Speed [m/s]	25
M2 Takeoff Weight [kg]	7.3	Wingspan [m]	1.524
M3 Takeoff Weight [kg]	5.0	Planform Area [m ²]	0.3618
Top Speed [m/s]	40	Banner Length x Height [m]	0.50 x 0.25
Number of Passengers	21	Estimated Score	5.82

2. Management Summary

2.1 Organization

To facilitate efficient communication and work, WUDBF is organized into an Executive Team, Advisors, and five Design Divisions. Team information and decisions flow downward from the Executive Team to Division Leads and ultimately to the General Members as represented by the hierarchical chart in Fig. 2.1. Throughout the design process new ideas are shared

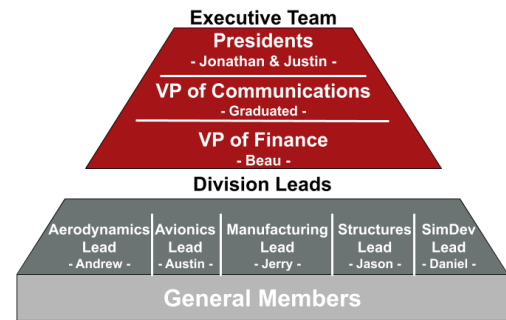


Figure 2.1: Team Hierarchy

throughout all levels of the team. The Executive Team consists of a Technical President, an Administrative President, a Vice President of Communications, and a Vice President of Finance. Executive Team members are responsible for aircraft technical decisions, administrative tasks, resource allocation, and public relations. Additionally, WUDBF is supported by two categories of advisors: graduate and undergraduate. Graduate Advisors provide industry and/or graduate level expertise and guidance to the team. Undergraduate Advisors are defined as any student who was previously a division lead or on the executive team and they counsel current team leads. There are five divisions (described in Table 2.1) and each is essential to the design and construction of the *Atlas*. The number of members assigned to each division is based primarily upon student interest, but also dependent on team needs.

Table 2.1: Relevant roles, assignments, and skills of WUDBF division personnel. Membership of each division is shown in parentheses.

	Aerodynamics (7)	Structures (4)	Manufacturing (4)	Avionics (3)	SimDev (2)
Roles	- Specify outer mold line of the aircraft - Ensure aircraft stability performance goals - Perform wind tunnel tests	- Design airframe - Analyze structure - Manage weight - Create drawings for manufacturing	- Manufacture from CAD - Select materials	- Design propulsion systems - Integrate control surfaces - Analyze telemetry	- Create and maintain simulation software - Analyze simulation results
Skills	- Aircraft stability, control, optimization - Numerical/CFD analysis	- CAD - Structural FEA - Mass properties analysis	- Laser cutting, 3D printing, machining - RC aircraft construction	- Propulsion theory - Electrical design - Soldering and crimping	- MATLAB/Simulink - Simulation - Real Flight 8 - Software debugging

2.2 Schedule

WUDBF follows a detailed schedule, shown in Fig. 2.2, in order to complete deadlines on time. WUDBF stayed on schedule during critical design stages and adapted to manufacturing and testing challenges to complete the *Atlas* on time.

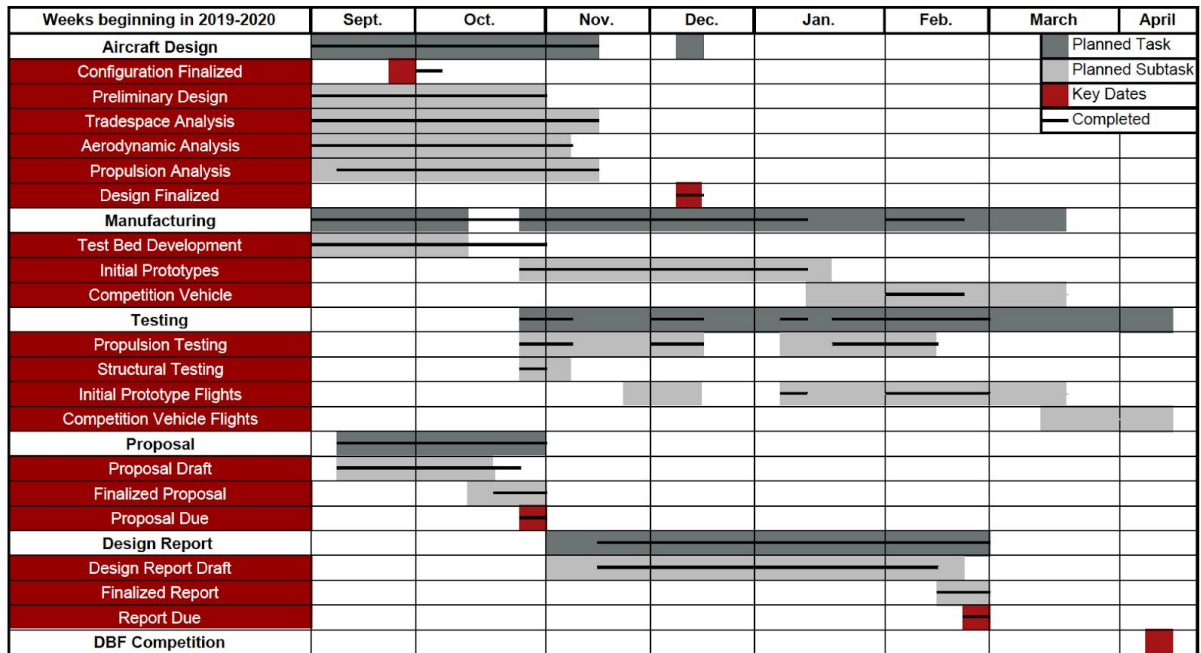


Figure 2.2: Gantt chart of WUDBF's design and build timeline for the *Atlas*.

3. Conceptual Design

3.1 Design Philosophy

The conceptual design phase begins with analysis of competition rules, mission requirements, and scoring formulas. The team determines the aspects which most influence the final score of the aircraft, such as lap time, passengers carried, and banner length. The most important capabilities are then translated into design requirements and goals. These goals are used to guide the selection of possible aircraft configurations, which are analyzed by the team. These aircraft are then modeled in XFLR5 to fulfil longitudinal and lateral-directional stability requirements. Finally, the aircraft are simulated for performance and score using WUDBF's in-house mission simulation models. Due to historic trends of adverse weather in Wichita, WUDBF emphasized risk reduction and aircraft handling qualities to maximize expected score (the analysis factors the probability of configurations successfully completing all missions in combination with predicted scores). The aircraft configurations are compared on predicted scores and perceived risk, guiding final configuration selection. The team selected the Pylon Racer configuration, christened the *Atlas*.

The 2020 DBF competition's mission requirements are based on real-world design challenges. This year, aircraft are required to tow a banner and carry passengers with luggage. These capabilities will be tested during four missions, detailed in section 3.2.

3.2 Mission Requirements

Mission 1 (M1), Mission 2 (M2), Mission 3 (M3), and the Ground Mission (GM) will be completed in accordance with the DBF rules and regulations. The team score is determined by Eq. 3.1.

$$\text{Overall Score} = \text{Written Report Score} * \text{Total Mission Score} \quad (3.1)$$

Where *Total Mission Score* is determined by Eq. 3.2

$$\text{Total Mission Score} = M1 + M2 + M3 + GM \quad (3.2)$$

A sample mission path is detailed in Fig. 3.1.

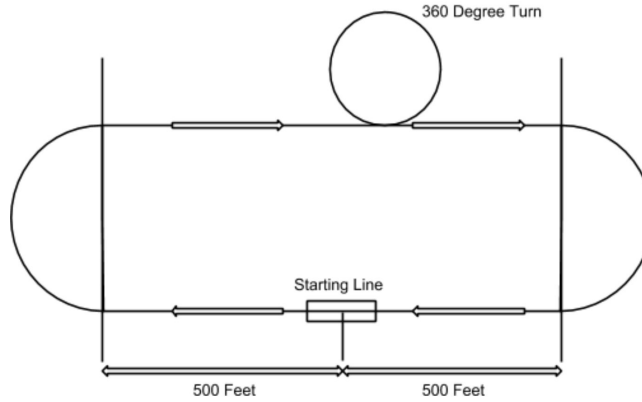


Figure 3.1: AIAA 2019 DBF competition course layout.

3.2.1 Mission 1

M1 is the test flight. The vehicle will begin the mission in its flight configuration without payload. It will take off within 20 ft. Once the aircraft throttle advances, it will have 5 min to complete three laps. The mission is completed with a successful landing. M1 is scored according to Eq. 3.3.

$$M1 = 1 \quad (3.3)$$

3.2.2 Mission 2

M2 is the charter flight. The vehicle enters the staging box in its flight configuration without payload. The payload for M2 consists of passengers and luggage. The assembly crew member will manually load the payload within 5 minutes. The vehicle will complete three laps within 5 minutes. After this, the pilot will successfully land the aircraft. M2 is scored according to Eq. 3.4.

$$M2 = 1 + [N_{\text{\#passengers/time}} / \text{Max}_{\text{\#passengers/time}}] \quad (3.4)$$

Where $N_{\text{\#passengers/time}}$ is the number of passengers carried by the *Atlas* divided by its mission time and $\text{Max}_{\text{\#passengers/time}}$ is the highest ratio of number of passengers carried to mission time among all teams.

3.2.3 Mission 3

M3 is the banner flight. The vehicle again begins in its flight configuration without the payload. A deployable banner with the *Washington University in St. Louis* logo will be loaded by the assembly crew member within 5 minutes. The vehicle will take off within 20 ft. The banner will remotely deploy on the downwind leg, after the first upwind turn, and remotely release on the last lap after crossing the finish line. The vehicle will have 10 minutes to complete as many laps as possible and then successfully land. M3 is scored according to Eq. 3.5.

$$M3 = 2 + [N_{\text{\#laps * banner length}} / \text{Max}_{\text{\#laps * banner length}}] \quad (3.5)$$

Where $N_{\#laps * banner\ length}$ is the product of the number of laps flown by the *Atlas* and banner length and $Max_{\#laps * banner\ length}$ is the highest product of scoring laps flown and banner length among all teams.

3.2.4 Ground Mission

The GM is timed and must be completed prior to M2 and M3. The vehicle will begin in a 10 ft square “mission box” with no installed payloads. The assembly crew member will enter the box, starting the timer, install the passengers and luggage, and exit the box pausing the timer. The pilot will demonstrate flight control functionality. Next, the crew member will return to the aircraft resuming the timer. They will remove the passengers and luggage, install the banner, and leave the box, stopping time. The pilot will prove the flight controls remain active. The plane will be held vertically and banner deployment will be demonstrated. The GM is scored according to Eq. 3.6.

$$GM = Min_{time} / N_{time} \quad (3.6)$$

Where N_{time} is WUDBF's time and Min_{time} is the minimum time among all teams.

3.3 Design Requirements

M1 requires aircraft to take off within 20 ft and meet minimum range requirements. M1 demonstrates mission requirements that must be exceeded to perform optimally during M2 and M3. M1 has negligible impact on overall score as it is fixed. The GM does not restrict aircraft design. All tasks completed in the GM must either be completed in flight or within the staging area before flight during M3 and M2, respectively.

To maximize M2 score, aircraft velocity and payload capacity must be maximized. This results in a model with high dynamic thrust to quickly complete three laps, and a large internal volume to store numerous passengers and luggage. There is no takeoff length limit in M2. However, the high dynamic thrust setup and additional structure required to store the internal passenger payload severely hinders take off performance required for M1 and M3. Changing propellers between missions is insufficient to resolve the takeoff condition. Limiting battery capacity to 200 Wh further limits takeoff capability. These conditions effectively place an upper bound on dynamic thrust. While WUDBF can achieve a high score by designing an aircraft primarily for M2 (see section 3.4.2), it cannot fully optimize for M2 without failing M1 takeoff requirements and, by extension, M3 qualification.

M3 prioritizes endurance and high velocity while requiring sufficient static thrust to take off in 20 ft. This drives WUDBF toward an efficient, high aspect ratio (AR) wing with low induced drag and a smaller body to minimize parasitic drag. Due to wingspan restrictions, higher AR wings limit wing area, increasing take off distance and serving as an AR limit. Optimizing for M3 decreases M2 score as interior storage is reduced for exterior banner capacity. Additionally, larger batteries are required to increase endurance, decreasing internal storage capacity. Optimizing for a single mission decreases expected score as risk increases. Since M2 and M3 prioritize different parameters, expected score has a local maximum where M2 and M3 configurations are co-optimized.

The optimal design balances M2 and M3, which also better distributes risk, increasing expected score. As a result, the Avionics Division must balance motor requirements between dynamic and static thrust, accounting for variable propeller diameter and battery size. Aerodynamics must balance low in flight drag (generally correlated with lower operating C_L) and high take off C_L , leading to the development of retractable high lift devices. Additionally, these design constraints push Aerodynamics and Structures to design an aircraft that has high internal volume for passengers and luggage and ample exterior mounting space for a banner. Summing these various design constraints, WUDBF produced several prototypes to explore these optimization constraints and compete for supremacy.

3.4 Initial Aircraft Designs

After preliminary analysis, design requirements yielded six aircraft designs that could surpass the mission requirements. Each aircraft was designed in XFLR5 and examined a different geometry, propulsion, avionics, and passenger count. All six aircraft met basic aircraft stability requirements. The conventional configurations used NACA 2412 airfoils and the deltas NACA 0010 for their realistic $C_{L,0}$ and well defined aerodynamic characteristics. These aircraft are presented in Fig. 3.2.

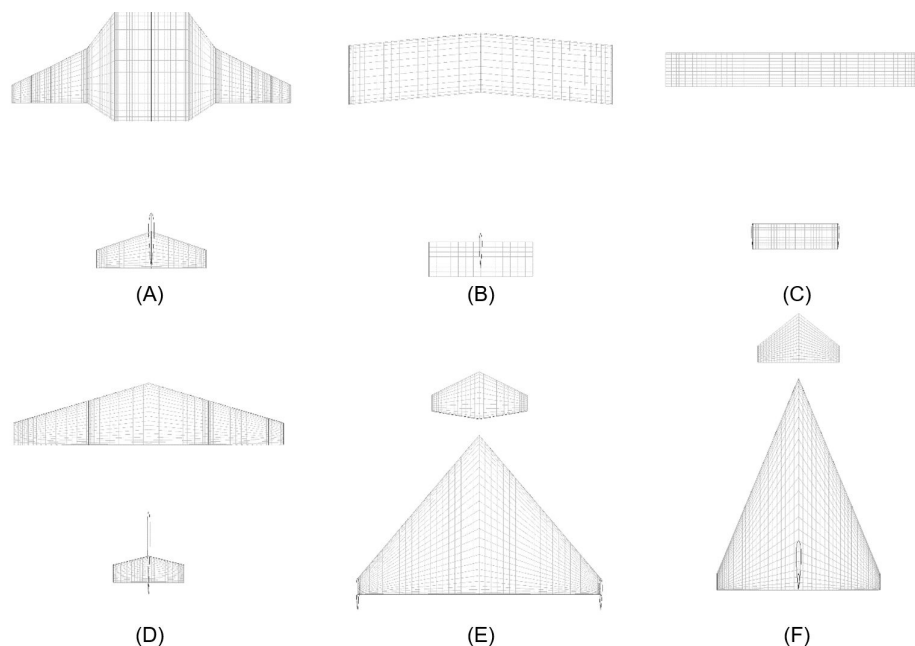


Figure 3.2: Initial aircraft planforms. A: Blended Body, B: Cargo Aircraft, C: Twin Boom, D: Pylon Racer, E: Delta Canard 5 Ft, and F: Delta Canard 3 Ft.

3.4.1 Blended Body

The Blended Body is a blended wing body, conventional-tail configuration that began as a flying wing utilizing a reflexed airfoil. The benefits of the flying wing were outweighed by the complexity of the design, so an empennage was added while preserving the blended, airfoil-shaped body. This design offers flexibility between tractor and pusher motors, and between single and twin motor configurations. Significant portions of the wing can be flapped for low-speed flight and the interior volume of the body

provides a high degree of flexibility for payload size and configuration. The main drawback of the design is manufacturing difficulties, specifically with regard to stiffness in the “body” section.

3.4.2 Cargo Aircraft

The Cargo Aircraft is a high-wing, conventional-tail configuration that sacrifices takeoff and cruise performance for maximizing payloads. It is designed to excel in M2, holding up to 30 passengers and luggage, at the cost of M3 score. The Cargo Aircraft’s wing is large to maximize lift and swept back to increase longitudinal stability and place the center of gravity (CG) over the main payload compartment. This entails a large fuselage volume for passengers and cargo that is still capable of fulfilling the short takeoff requirement. The Cargo Aircraft is somewhat flexible in motor configuration — it can support both a single and twin tractor motors. High lift device integration is simpler on a conventional wing.

3.4.3 Twin Boom

The Twin Boom is a high-wing, H-tail configuration that was fully designed as last year’s competition vehicle and updated for this year. It highlights the benefits of having multiple fuselages at the costs of additional parasitic drag. This particular twin-boom aircraft has a twin puller motor configuration and does not employ the thrust-vectoring seen in WUDBF’s 2019 aircraft. Instead, it only uses blown trailing edge flaps for low speed flight. Having two separate fuselages presents manufacturing difficulties of rigidity and weight distribution that are otherwise nonissues.

3.4.4 Pylon Racer

The Pylon Racer is a low-wing, T-tail aircraft based on the class of aircraft by the same name built for high speed maneuverability. These aircraft benefit from lower wing loading, which decreases maximum payload. It has a single tractor motor configuration and a high degree of flexibility for high lift devices. Pylon racing aircraft are not designed for high payload capacities, requiring increased wing area and wing loading. One key concern of this is maneuverability, necessitating additional analysis.

3.4.5 Delta Canard 5 Ft

The Delta Canard 5 Ft is a delta-wing canard, twin-tail configuration with a 5 ft wingspan. This necessitates a large wing area, with rigidity becoming a major manufacturing concern. Despite this, the configuration is favorable due to high maneuverability at high angles of attack (AoA). Single or twin motors must be pushers to not disturb the delta airflow. However, this eliminates the possibility of blown flaps which, in conjunction with the delta wing’s poor takeoff performance, represents a major drawback for takeoff weight. This is the main subject of further analysis for this configuration.

3.4.6 Delta Canard 3 Ft

The Delta Canard 3 Ft is a delta-wing canard, conventional-tail configuration which accounts for the rigidity concerns of the Delta Canard 5 Ft and is specialized for high-speed, low-capacity flight. This is the only aircraft that has a wingspan of less than 5 ft. The 3 ft wingspan ensures manufacturing concerns are manageable. Performance concerns are not, however, solved by this model, necessitating a lower payload capacity. This aircraft’s performance determines whether a low capacity, high speed

configuration is preferable over a high capacity, low speed configuration. The motor configuration is unchanged from the Delta Canard 5 Ft.

3.4.7 Configuration Qualitative Comparison

Table 3.1 presents a summary of the qualitative differences between configurations. A “+” implies advantages, a “-” disadvantages, and “=” means that the compared categories are similar.

Table 3.1: Qualitative differences between considered aircraft configurations.

Category	Blended Body	Cargo Aircraft	Twin Boom	Pylon Racer	Delta Canard 5 Ft	Delta Canard 3 Ft
M2 Performance	=	+	=	=	-	-
M3 Performance	+	-	=	=	+	+
Manufacturability	=	+	=	+	-	=
Propulsion Flexibility	+	=	+	-	-	-

3.5 Final Design Selection

3.5.1 Initial Configuration Scoring Analysis

To guide team design decisions, each aircraft flew simulated missions in KM Sim to produce performance and mission scoring estimates. KM Sim is a MATLAB-based idealized control dynamics simulation of an aircraft using a prescribed course. Aircraft aerodynamic properties, inertial properties, and mission-specific parameters were manually compiled for each airplane. The simulator uses configuration specific data to model relevant states during each mission. The aircraft aerodynamics and propulsion models are table-lookup values from XFLR5 and the University of Illinois at Urbana-Champaign propeller database, respectively.

Simulated flight phases in KM Sim include ground roll, rotation, optimum climb, max climb, cruise, constant radius and constant g-load turns, and optimum glide. Each flight phase is modeled as a separate system of differential equations. It is numerically integrated, producing a time history of 14 states, including, velocity, orientation, battery charge, and other useful data. This time history is used for data visualization and scoring. A sample trajectory plot, shown in Fig. 3.3, displays the complete flight path of an aircraft using position and flight phase data. Flight paths are plotted within an East-North-Up coordinate space.

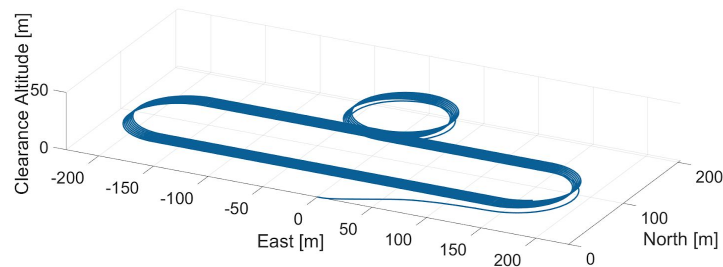


Figure 3.3: Trajectory plot of KM Sim results.

KM Sim also produces graphs of important states, shown in Fig. 3.4, as an eight-panel results summary of the flight. The flight summary contains time history plots of the most useful states. These data are useful for quick visual analysis to make sure the simulated flight was performing as required. Some examples include checking maximum and nominal values of AoA, the mission phase in case of a failed flight, and remaining battery charge.

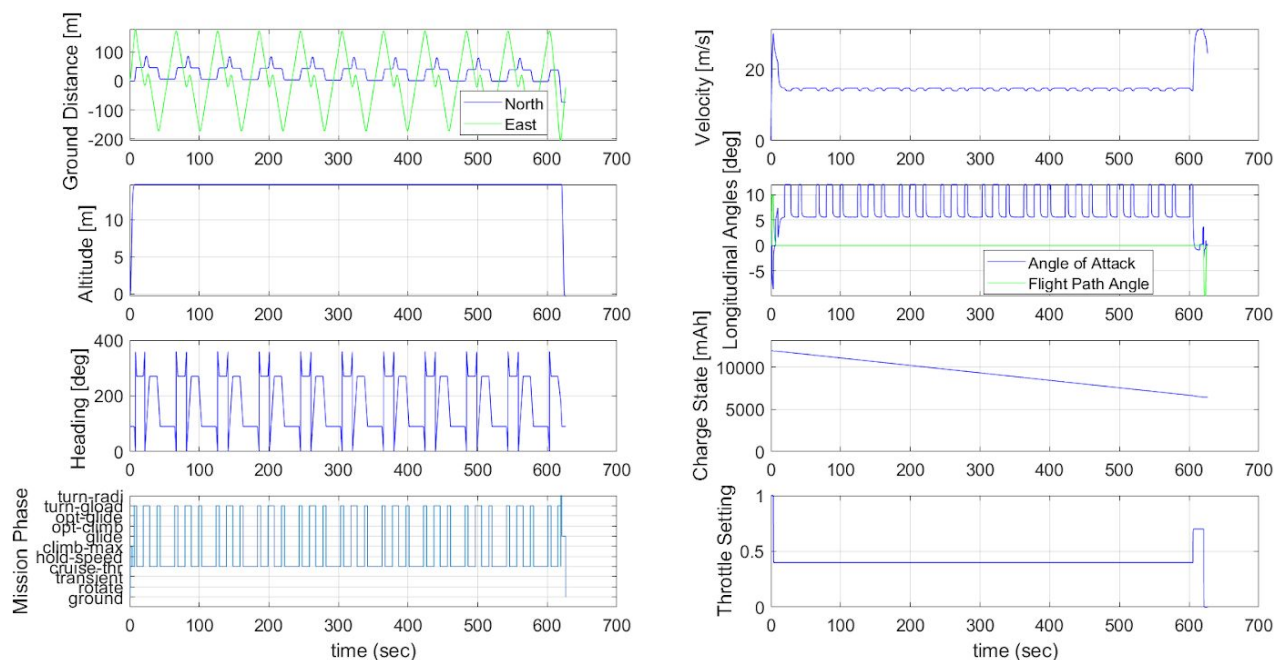


Figure 3.4: Example of eight panel results summary.

KM Sim has the necessary features to simulate each of this year's missions. For example, KM Sim alerts the user when the aircraft did not takeoff within 20 ft, but continues the simulation for analysis of the rest of the flight. It also supports aircraft configuration changes, like flaps, and loadout changes, such as unfurling the banner. Configuration and loadout changes are performed by switching or modifying the aerodynamic lookup tables. KM Sim can be employed for a wide variety of studies. For instance, aircraft designs can be run through different courses with the same initial state or through the same course with different payload, flap, and thrust vectoring instructions. Additionally, parameter studies, range calculations, and maximum takeoff weight calculations are all easily performed in KM Sim.

3.5.2 Sensitivity Analysis

Due to competition limitations on wingspan, the aircraft AR is strongly correlated to the planform area of the aircraft. The additional efficiency of a high AR wing must be balanced with the additional lift of a larger, low AR wing. The short takeoff requirements of the competition slightly favor a larger wing area at the cost of efficiency and the sensitivity study in Fig. 3.6 demonstrates this; the Blended Body performs best due to its maximized main wing area. A partial derivative value of -0.043 score/AR was found in the analysis and the final *Atlas* design had an AR of 6.4.

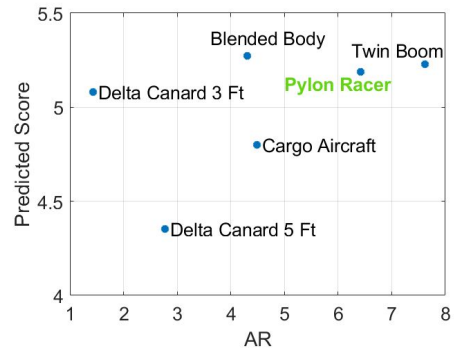


Figure 3.5: Scoring plot of estimated total score and aspect ratio.

The second sensitivity study entails battery mass; a larger mass increases either power or endurance but impairs wing loading and, for M3, takeoff distance (see section 4.4.1). The battery mass partial is found via the chain rule from partials with respect to voltage and charge capacity, and reveals that increasing battery mass has a positive (albeit small) impact on score. Since the chosen battery mass lays at the upper limit of allowed energy (and thus mass), the battery optimizes score for this configuration. The analysis yielded a value of 0.003 score/kg for the battery mass.

3.5.3 Aircraft Risk Analysis

To execute the team's strategy of maximizing the most probable score, aircraft design risk must be reduced. WUDBF rated each of the aircraft on five risk-related categories to quantify each design's overall risk. The CAD score represents the difficulty to create the aircraft within SolidWorks. The Make score captures the complexity of sourcing raw materials and creating the aircraft components. Assembly reflects the difficulty of assembling all of the aircraft's components. Structure refers to the difficulty of making the configuration structurally sound. Finally, Stability embodies the difficulty of ensuring longitudinal and lateral stability as well as favorable handling qualities. This risk matrix is shown in Table 3.2 and then plotted with expected score in Fig. 3.5. Note that the two Delta Canard configurations were given the same risk scores due to their similarity.

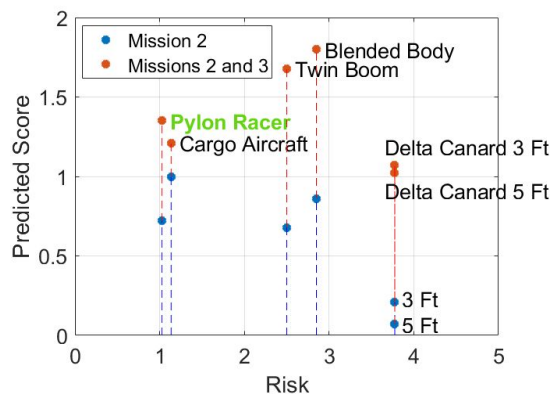
Risk categories were given weights to reflect their severity, with higher weights reflecting more serious risk. The CAD risk category was weighted at 10% because the difficulty in modelling an aircraft is largely independent of configuration. The Make category was given a weight of 25% because component sourcing and fabrication may be out of the team's control. A design that requires expensive materials and components that require specialized tools represents risk that cannot be overcome as easily as risk associated with other categories. Assembly risk was weighted at 15% due to the team's historical struggles with aircraft manufacturing and the decision to work with wood, with which the team has little experience. Structure and Stability were each weighted at 25% because they represented the viability of the configuration itself and were not dependent on the team. Thus Make, Structure, and Stability risks are inherently more risky than CAD and Assembly risks.

Table 3.2: Aircraft configuration risk matrix.

Risk Category	Weight (%)	Cargo Aircraft		Blended Body		Delta Canard		Pylon Racer		Twin Boom	
		Rating	Weighted	Rating	Weighted	Rating	Weighted	Rating	Weighted	Rating	Weighted
CAD	10	1.1	0.1	3.5	0.4	3	0.3	1	0.1	2.5	0.3
Make	25	1.5	0.4	3	0.8	5	1.3	1	0.3	3	0.8
Assembly	15	1	0.2	2	0.3	4	0.6	1	0.2	3.5	0.5
Structure	25	1	0.3	3	0.8	4	1	1	0.3	3	0.8
Stability	25	1	0.3	2.8	0.7	2.5	0.6	1.1	0.3	2.5	0.6
Total score		1.3		3.0		3.8		1.2		3.0	
Rank		2		3		5		1		4	

The Cargo Aircraft and Pylon Racer have low risk scores due to a simpler wing and fuselage design. Further, their smaller wings simplify manufacturing. They can also easily be made strong and stable. The Blended Body has high risk scores due to the anticipated complexity of the airframe, the more complicated components, and the aerodynamic characteristics simulated by the Aerodynamics Division. The Delta Canards have extremely high risk scores because of the large size of the wing. The team estimated that it would be costly and difficult to source wood and other components large enough for the wing ribs and furthermore did not have tools large enough. As a result, the Manufacturing Division determined that the Delta Canard would be nearly impossible to make. The Twin Boom configuration has a medium-high risk score due to anticipated structural challenges in making the plane rigid given the high moment on the booms connecting the tail to the fuselage. These rigidity issues also represent stability risks, resulting in a medium-high stability risk score.

Each aircraft is plotted for its normalized M2, M3, and weighted risk, shown in Fig. 3.5. The score is calculated assuming that each aircraft flies all three missions, where only M2 and M3 are displayed as performance in M2 and M3 determines what additional score.

**Figure 3.6: Scoring plot of estimated M2, M3, and measured risk.**

Several aircraft converged to a final score within a half of a point of one another at the top of the graph. WUDBF interpreted this result as a demonstration of score independence relative to configuration.

As a result, the configuration choice could be made based on factors beyond performance, such as manufacturability.

3.5.4 Final Configuration Decision

The final aircraft configuration of the *Atlas* was informed by the analyses of the six initial designs.

The *Atlas* most closely resembles the Pylon Racer configuration due to its high predicted score and minimal associated risk. The *Atlas* configuration has a high-wing with a swept leading edge, a straight trailing edge, and a conventional tail. The straight trailing edge ensures that the high lift devices do not need to be mounted at an angle, reducing aerodynamic concerns of spanwise flow. The leading edge is swept to achieve a taper ratio of 0.4 to minimize induced drag. The high-wing eliminates the need for dihedral and provides greater ground clearance for the trailing edge high lift devices and propellers. The conventional tail was chosen to minimize the necessary structure in the vertical stabilizer.

4. Preliminary Design

4.1 Aerodynamic Performance

The *Atlas* design was iteratively improved to ensure the plane is stable, controllable, and achieves maximum aerodynamic performance. Both XFLR5 and MATLAB were used for aircraft analysis. The size and shape of the main wing and stabilizers were driven by iterative static and dynamic stability analyses. Control surfaces and high lift devices were dimensioned to improve handling characteristics and minimize takeoff distance, respectively.

4.1.1 Main Wing Planform Area

To maximize lifting capacity, the wingspan is 5 ft, the maximum allowed. A manufacturing constraint was a straight trailing edge for the trailing edge Fowler flaps. Finally, the performance goals were to cruise at 30 m/s near 0° AoA while having a stall speed of 8 m/s with flaps extended. Given these constraints and a desired taper ratio of 0.4, a wing area of 0.36 m² was selected, yielding 58 N of lift at cruise, a cubic wing loading (CWL) factor of 21 oz/ft³, and an AR of 6.4.

4.1.2 Wing Design for Aerodynamic Performance

Span limited aircraft with an elliptical lift distribution have minimal induced drag; this distribution is common in most conventional aircraft [1, 2]. For the main wing, the Aerodynamics Division applied twist and tapering to approximate an elliptical lift distribution. Along the span, the twist was applied at four locations, shown in Table 4.1. Minimizing these locations aids in manufacturing. The actual lift distribution closely models the ideal elliptical shape, shown in Fig. 4.1. Additionally, by de-twisting the wing at the wingtips, the aircraft maintains roll control in a partial stall and tip-stall tendencies are weakened.

Table 4.1: Main wing twist values as a function of distance from the root.

Distance from root [mm]	Twist [°]
0	0
339	0.5
677	0
762	-4.0

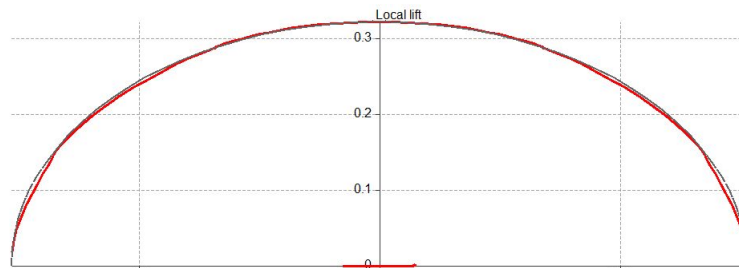


Figure 4.1: Elliptical ideal [black] and actual [red] lift distribution of main wing at 0.74° AoA.

4.1.3 Main Wing Airfoil Selection

The Reynolds number (Re) significantly reduced the scope of the main wing airfoil selection. The predicted Re of the plane was calculated, using Eq. 4.1.

$$Re \equiv \frac{\bar{c} V_{\infty}}{\nu} = \frac{0.237 \cdot 30}{1.498 \cdot 10^{-5}} = 474\,633 \quad (4.1)$$

where $\bar{c}$ is the mean aerodynamic chord [m], V_{∞} is the freestream velocity [m/s], and ν is the kinematic viscosity of air [m²/s]. Using XFLR5, kinematic viscosity was calculated from the average temperature in April (14 °C) and altitude (400 m) of Wichita. The team searched the University of Illinois at Urbana-Champaign database for airfoils with a maximum C_L/C_D ratio at the cruise Re at AoA between 0° and 2° which also performed well at low Re and high AoA. In the search parameters, the maximum thickness was set to 12% to use thin airfoil theory approximations. Additionally, a minimum thickness was set to 8% for structure and system integration. The maximum camber was set to 4% to minimize drag while maintaining high lift. The results returned three potential airfoils as shown in Table 4.2.

Table 4.2: Final airfoil selection.

	Max. C_L/C_D at $Re = 500\,000$	Max. Thickness [% of chord]	Max. Camber [% of chord]
Eppler E193	114	10.2	3.0
Eppler E195	112	11.8	2.7
Clark Y	98.7	11.7	3.4

The Eppler E193 was chosen since it has the highest C_L/C_D at the cruise Re and performs well at low Re and high AoA, all while maintaining a high level of manufacturability. Additionally, WUDBF has experience with Eppler airfoils. Figure 4.2 shows the Eppler E193 airfoil.

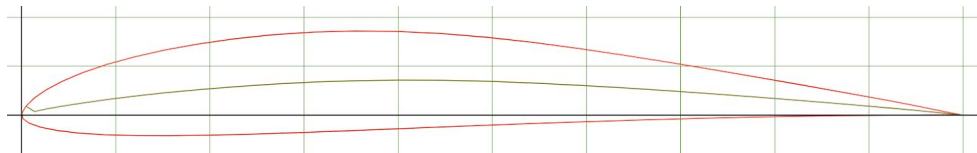


Figure 4.2: A cross section of the chosen Eppler E193 airfoil.

4.1.4 Stabilizer Airfoil Selection

Stabilizer airfoils are generally symmetric as they may either provide upforce or downforce, depending on the final trim state of the aircraft. The team selected the NACA 0010 airfoil as it is common for stabilizers and used successfully in past WUDBF projects.

4.1.5 High Lift Devices

WUDBF employs a series of high lift devices to generate enough lift on takeoff. The high lift devices decrease takeoff speed and takeoff length benefiting M1 and M3 performance. Additionally, WUDBF has knowledge on Fowler flaps, which were implemented to decrease takeoff length in last year's aircraft. The theoretical additional lift values for the Fowler flaps are from published data [3]. The Fowler flaps are placed in the propwash of the motors, increasing the relative freestream air speed over the flaps, increasing their effectiveness in low-speed, high-power flight. The flap has a chord fraction of 0.3, a span of 350 mm, and deploy at a 39° angle. The increment of section maximum lift coefficient at 39°-deflection is $\Delta C_{L_{max}} = 1.75$ for the Fowler flap, which is greater than either of the other slotted flap options it is compared against [3]. A "flap fusion" mode was also implemented where the ailerons deploy downward as well, increasing the total deployed area of the wing.

Leading edge high lift devices were considered because they further decrease takeoff speed by increasing maximum AoA and thus, the maximum C_L . Specifically, Krueger flaps offer the best compromise between high lift performance and weight. Krueger flaps were designed with published guidelines [4]. The design parameters were a rotation angle of 130°, a flap length of 20% of the chord, and a flap nose thickness of 5 mm [4]. They were ultimately removed to streamline manufacturing and maintain a structural leading edge.

Winglets will be implemented in the next iteration to increase lift performance and were tested at various speeds in viscous and inviscid analyses using XFLR5. The additional lift generated as a function of winglet area hits a plateau beyond a certain point; the area was therefore chosen as the point at which doubling the winglet area only led to a 10% increase in additional lift: 94 cm² per winglet. The winglets are upward facing for ground clearance. The Manufacturing and Structures Divisions determined that this design is feasible and the additional lift generated outweighs increased manufacturing complexity.

4.1.6 Control Surface Sizing

Given the traditional design of this aircraft, all control surfaces (rudder, elevator, and ailerons) were designed using basic moment equations and rule-of-thumb parameters proven on past aircraft. The moment balance equations are Eq. 4.2 through Eq. 4.4

$$L_p p + L_{control} = 0 \quad (4.2)$$

$$M_q q + M_{control} = 0 \quad (4.3)$$

$$N_r r + N_{control} = 0 \quad (4.4)$$

The lateral and longitudinal derivatives were taken from an XFLR5 dynamic stability analysis. The "control" moments were also generated in XFLR5 using deflections of the control surfaces at varying speeds. The control surface sizes on the model were tweaked until the rates met the pilot's standards. Chord fractions for the control surfaces on the first prototype were 40% for the ailerons, 30% for the elevator, and 20% for the rudder. These control surface sizes were validated in flight. The pilot listed the

initial aircraft as a 3 on the Cooper-Harper scale, indicating satisfactory but suboptimal flight characteristics. The aircraft was slightly too responsive in roll while greater pitch and yaw authority would increase pilot comfort. The rudder and elevator were enlarged on the next iteration in response.

4.2 Model Capabilities and Uncertainties

KM Sim is a simulator of ideal aircraft performance that is useful for initial configuration choice and basic mission performance modeling, but cannot be used as a high-fidelity simulator. The flight path is preplanned and does not take into account wind gusts or pilot control. The initial aerodynamic calculations are based on table look-ups of pre-calculated data from XFLR5, which poorly models parasitic drag. However, KM Sim can take in data from any source in table format, allowing for improved modelling from wind tunnel data (see section 8.4). Aircraft surface areas are modeled as flat plates to estimate parasitic drag. KM Sim provides an estimate of flight performance metrics (Fig. 3.3). The results allow the team to remove unviable configurations, making final selections easier.

After configuration selection, KM Sim was upgraded with empirical data from flight tests, motor tests, and wind tunnel tests. Flight test data provided the required throttle setting to fly at cruise speed, which helped correct inaccurate drag calculations. Motor test stand data provided the correct maximum thrust value and the throttle breakpoints values. Finally, wind tunnel test data provided more accurate C_L data due to features not modeled in XFLR5. With these additional data sources, KM Sim outputs closely matched flight test data (see section 8.5), shown in Fig. 4.3. The left plot represents the straight section of flight chosen from the flight test, and the right plot represents the simulated flight using the updated KM Sim. From visual inspection during the test flight, the *Atlas* flew at an AoA range of 0-2°, confirmed by KM Sim.

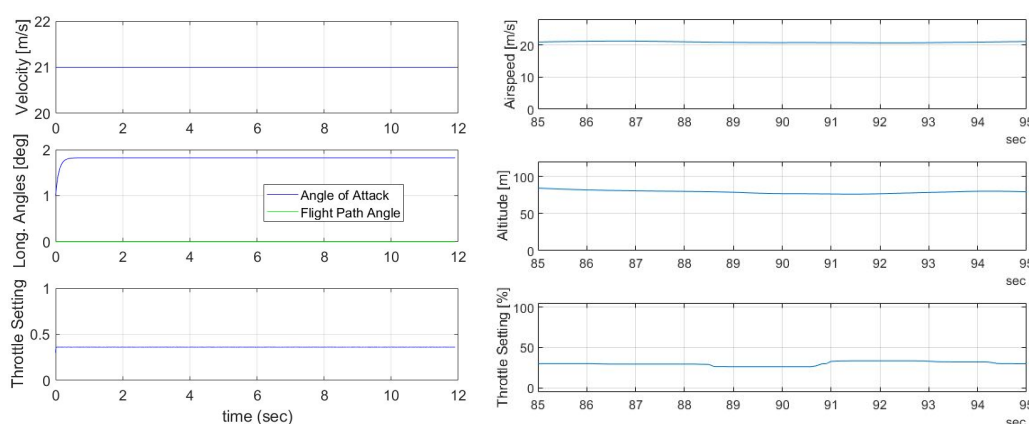


Figure 4.3: Left: KM Sim predicted performance. Right: flight test data.

4.3 Aerodynamic Analysis

4.3.1 Longitudinal Stability and Horizontal Stabilizer Sizing

Longitudinal stability is fundamental to basic aircraft design and is a function of the location of the center of mass, the added lift provided by the main wing, and the added lift provided by the horizontal

stabilizer. The center of mass is liable to shift between missions with the addition of passengers and cargo or a banner. Ensuring that the aircraft's longitudinal stability falls within the desired range for all loading configurations is vital for a successful mission and competition. Longitudinal stability is dependent on several factors which are shown in Eq. 4.5 [5].

$$C_{m_\alpha} \equiv \frac{\partial}{\partial \alpha} C_m = C_{L_{\alpha_w}} \frac{l_{ac_w}}{\bar{c}} + C_{L_{\alpha_h}} \left(1 + \frac{\partial \varepsilon}{\partial \alpha}\right) \eta_h \frac{S_h l_{ac_h}}{S \bar{c}} \quad (4.5)$$

Equation 4.5 shows the distance from the aerodynamic centers of the main wing and horizontal stabilizer to the CG ($l_{ac,w}$, $l_{ac,h}$), and the size of the horizontal stabilizer S_h are variables that affect longitudinal stability. Elevator distance was driven by longitudinal eigenvalue analysis, seen in section 4.3.3. Based on prior aircraft designs of similar size and team experience, $C_{m,\alpha}$ values between -0.5 and -1.8 were appropriate. The $-1.05 C_{m,\alpha}$ at $C_m = 0$ fit within the bounds of the guidelines and is resilient to changes in configuration shifting the center of mass.

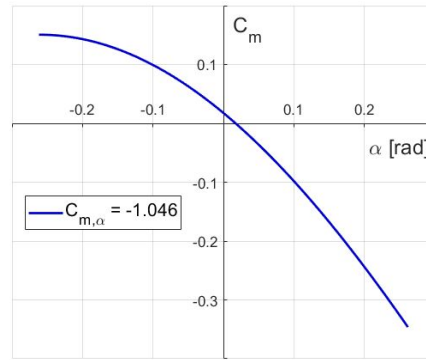


Figure 4.4: Longitudinal stability curve, C_m vs. α .

4.3.2 Directional and Lateral Static Stability

Directional and lateral stability modes were determined after completing longitudinal stability analysis. Since the lateral directional stability space links lateral and directional modes, individually looking at their static stability derivatives does not provide a complete view of the aircraft dynamics. An analysis of the dynamic stability poles was deemed more useful for understanding directional and lateral stability. Figure 4.5 shows the lateral and directional static stability slopes which are as follows: roll stability is negative with a $C_{l,\beta}$ of $-1.29\text{e-}3 \text{ rad}^{-1}$ while the directional stability, $C_{n,\beta}$, is $3.06\text{e-}3 \text{ rad}^{-1}$.

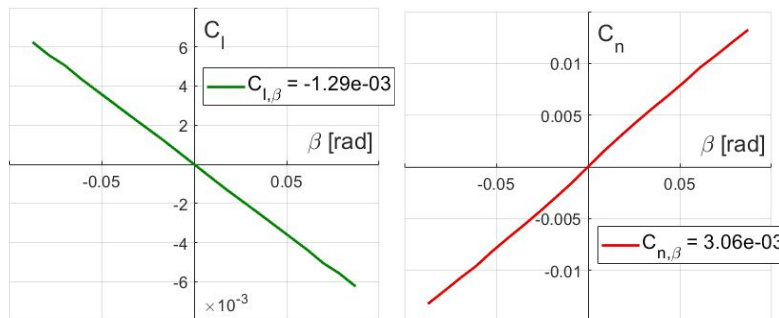


Figure 4.5: Left: Lateral stability curve, C_l vs. β . Right: Directional stability curve, C_n vs. β .

4.3.3 Longitudinal Eigenvalue Analysis

Longitudinal eigenvalue analyses conducted in XFLR5, shown in Fig. 4.6, drove horizontal stabilizer distance from the CG. When plotted on the real and imaginary axes, the *Atlas*'s complex roots for longitudinal stability case lie in the left (negative) half plane, which is necessary to ensure stability [5]. WUDBF chose to constrain analysis and iteration to the short period mode since the long period mode has a high enough time scale such that minor pilot control inputs can eliminate oscillations [5]. Iterating through a range of tail distances, WUDBF determined the proper tail distance by comparing the damped and undamped frequencies to those of other aircraft, shown in Fig. 4.7. The final shape of the longitudinal eigenvalues, shown in Fig. 4.6, is similar to other aircraft and serves as additional confirmation that aircraft dynamics have been predicted correctly [6].

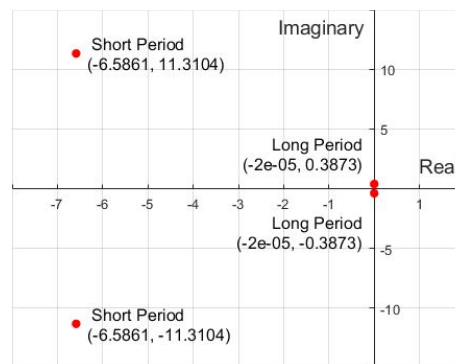


Figure 4.6: Longitudinal eigenvalues characteristic short and long period modes for final horizontal stabilizer.

Table 4.3 displays the short period and long period natural and damped frequencies obtained from the eigenvalue analysis in Fig. 4.6.

Table 4.3: Short period and long period natural and damped frequencies.

Mode	Natural Frequency [Hz]	Damped Frequency [Hz]	Damping Ratio
Short Period	2.08	1.80	0.503
Long Period	0.062	0.062	0.00051

The *Atlas*'s natural frequency and damping ratio of the short period mode were found to provide acceptable handling qualities at cruise using XFLR5's dynamic stability analysis tools. The damping ratio was satisfactory and the natural frequency, while large for full-size aircraft, is in line with other small UAVs [6]. The *Atlas*'s short period mode natural frequency plotted versus wingspan follows the trend of a higher natural short period frequency for a smaller aircraft, shown in Fig. 4.7 [6].

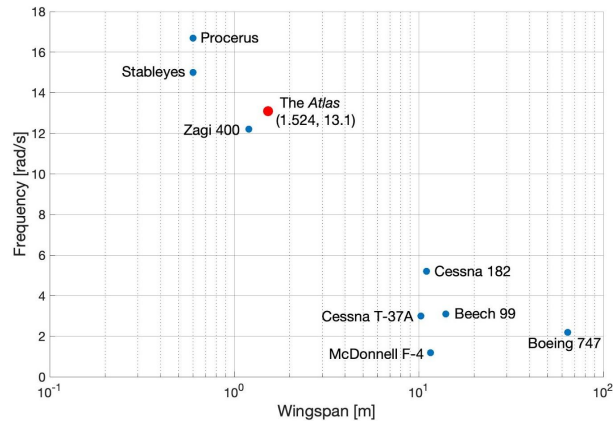


Figure 4.7: Natural short period frequency vs. wingspan for various aircraft [6].

4.3.4 Lateral-Directional Eigenvalue Analysis

The lateral-directional eigenvalue analysis determined vertical stabilizer size and position. The Dutch roll stability characteristics were compared to other aircraft in Fig. 4.9 to ensure the *Atlas*'s eigenvalues were in line with similar aircraft. As shown in Fig. 4.8 and Table 4.4, the spiral mode and roll mode time-to-halve are extremely large and small, respectively, which was expected. Having a high time-to-halve for spiral mode and low time-to-halve for roll mode are characteristics of a laterally stable aircraft, therefore no design changes were made based on these two roots [5]. The *Atlas*'s lateral-directional eigenvalues will produce dynamic stability in addition to satisfactory handling qualities.

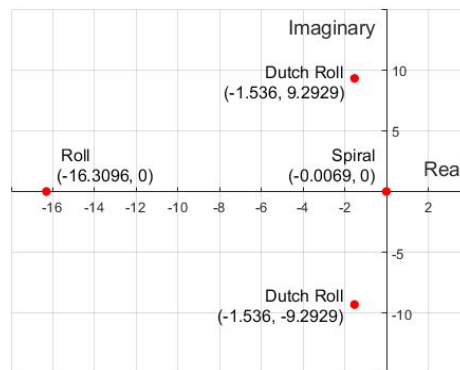


Figure 4.8 Lateral directional eigenvalues: roll root, spiral root, and Dutch root.

Table 4.4 displays the characteristics of the Dutch roll eigenvalue.

Table 4.4: Dutch roll, Spiral, and Roll Modes

Mode	Natural Frequency [Hz]	Damped Frequency [Hz]	Damping Ratio	Time to Halve [s]
Dutch Roll	1.06	1.05	0.141	-
Roll	-	-	-	100.1
Spiral	-	-	-	0.042

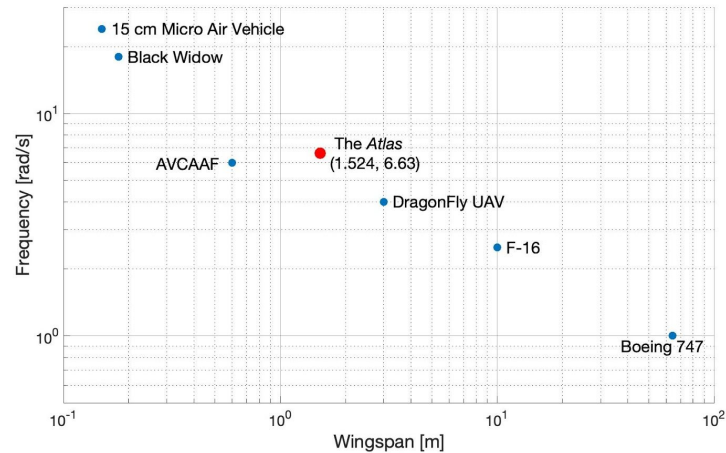


Figure 4.9: Dutch roll natural frequency vs. wingspan size overlaid with other aircraft [6].

Lateral stability poles for previous design iterations are shown in Fig. 4.10 as part of a sensitivity study of empennage sizing. The “Final” configuration is on the log-log trend of Fig. 4.9.

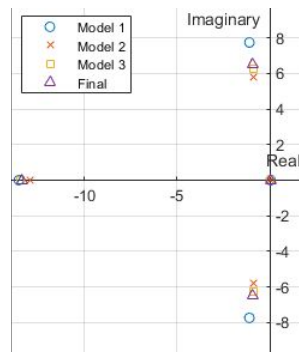


Figure 4.10: Lateral stability root locus plot with early iterations graphed.

4.3.5 Drag Determination

While XFRL5 provided accurate lift and induced drag measurements, the analysis considered only inviscid flow. Parasitic drag and drag due to body features were not included in the model. A parasitic drag script was created to compensate, modeling objects as thin plates. The thin-plate drag estimate was then corrected for thickness using wind tunnel data correlations (See Section 8.5).

4.3.6 Banner Analysis

Determining an exact value for banner drag at cruise proved experimentally and analytically difficult. The wind tunnel has no feasible banner mounting mechanism. Less controlled methods for empirically finding drag were pursued.

Candidate banners were designed using Ripstop nylon, fabric, foam board, Kapton tape, and fiberglass tape. Banners were attached to a spring scale and placed in a 30 mph freestream. Objective values were difficult to find due to the wide variation in drag forces arising from the banner's exact orientation, so average drag values for the banners were calculated and compared.

Initial results showed that a light, unstiffened fabric produces significantly less drag than a rigid or semi-rigid material. Drogue chutes and weights are yet to be analyzed for their ability to maintain an orientation. A test banner is shown in Fig. 4.11.



Figure 4.11: Low drag banner.

4.4 Propulsion Sizing

The propulsion system on the *Atlas* is designed to maximize expected total mission score by maximizing cruising speed while adhering to aerodynamic, structural, and takeoff requirements. A balance of good cruising speed and risk reduction is the most important consideration for maximizing the expected score (see section 3.3). Since cruising speed is directly dependent on thrust, the goal of the propulsion system is to maximize thrust at cruise speed, or dynamic thrust, with three key constraints:

1. Endurance: Design cruising efficiency to be high enough for the propulsion system to last the 10-minute time window for M3 with the Lithium Polymer (LiPo) battery energy constraints.
2. Handling Characteristics: Ensure low wing and power loading to provide good handling qualities and decrease risk.
3. Static thrust-to-weight ratio (T/W): Ensure static T/W is sufficient for 20 ft takeoff for M3 and M1.

4.4.1 Initial Sizing

An initial range of acceptable battery masses was first estimated from structural and aerodynamic requirements (section 4.1) as well as competition rules.

Aerodynamic calculations provide estimated aircraft empty from the desired CWL. CWL extends the concept of wing loading, the weight per unit of wing area (A), to be scalable to different aircraft sizes:

$$CWL = \frac{AUW}{A^{1.5}} \quad (4.6)$$

Where AUW is all up weight. CWL provides a rule-of-thumb characterization of aircraft handling qualities, regardless of aircraft size. Heavier aircraft, such as racers, tend to have a CWL range of 15-20 oz/ft³ [7], and the wing loading can be further increased while still controlling risk as long as the power loading, or aircraft weight per unit propulsive power, is reduced [8].

Competition rules encourage a high wing loading on M3 as designing for a higher energy, and therefore heavier battery, allows power output to be increased, providing more thrust to counteract banner drag and thus make the plane faster and carry a bigger banner. On M2, heavier motors allow a higher

discharge rate allowing the plane to fly faster. A higher wing loading therefore increases expected score up to some point, at which the risk of poor handling qualities and a low glide-ratio due to the high wing loading decrease expected score. On M3, a short takeoff distance of less than 20 ft bounds wing loading, but this limitation can be overcome with a higher thrust to weight ratio as well as efficient high-lift devices such as the Fowler flaps on the *Atlas*, so overall scoring encourages higher wing loadings.

The initial design of the *Atlas* therefore aimed to have a relatively high CWL at 20 oz/ft³ in M3, reasonable for the low power loading that can be achieved with a high specific energy LiPo chemistry battery. AUW and empty weight (EW), that is, the aircraft weight minus the payload, were then derived from this wing loading and a wing planform area of 0.36 m². This sets the desired EW at 4.5 kg, and maximum desired passenger capacity at 2.4 kg. If the plane handles poorly with all 21 passengers, passengers can be removed and the aircraft will still score well in M2 due to a high airspeed.

The aircraft fuel fraction is the ratio of battery weight to EW. Given EW estimates and typical DBF fuel fractions of (Table 4.5), the battery weight allowed by structural considerations was predicted.

Table 4.5: Historic DBF fuel fractions [9, 10, 11].

Aircraft	Fuel Fraction [%]
WUDBF	35-40
The Spirit 2018-19	37
Spruce Zeus 2017-18	35
Wrighton Flyer 2016-17	38
Winning DBF Teams	20-40
Ljubljana 2018-19	41 ¹
Clarkson 2017-18	38
USC 2016-17	22

Assuming a 35% fuel fraction, given past fuel fractions, reasonable battery masses for a 4.5 kg aircraft range from 1.35-1.8 kg. However, the 200 Wh LiPo battery pack total energy limit further constrains battery mass. High quality LiPo batteries have a specific energy of approximately 150 Wh/kg; the 200 Wh rule limits battery mass to 1.3 kg. WUDBF thus decided to use the maximum pack size within this limit: a 200 Wh, 1.3 kg battery pack. This gives the *Atlas* a fuel fraction of 28% and additional weight margin that can be devoted to motor mass (see section 4.4.3).

4.4.2 Motor Configuration

The *Atlas* can support both one- and two-motor configurations (see section 3.4.2), but analysis favors two. The motor configuration optimizes motor-propeller efficiency and propeller diameter. Disk actuator theory, which provides basic propulsive relationships for the static case, holds that [12]:

$$\text{Mechanical Power} = \sqrt{\frac{T^3}{2\rho A}} \quad (4.7)$$

¹ Thanks to the University of Ljubljana's DBF team for providing this information

where T is thrust [N], ρ is air density [kg/m³], A is total propeller disk area [m²], and mechanical power [W] is the propeller's rotational speed multiplied by torque at steady state. Thus, a larger total propeller area decreases the required power to obtain the same thrust, increasing static mechanical efficiency (thrust over mechanical power).

Achieving high static thrust (for 20 ft takeoff in M3) on a one-motor configuration therefore requires a large propeller diameter. For a multi-motor configuration, however, multiple smaller propellers reach similar efficiencies with the same *total* disk area. A large propeller diameter can actually be disadvantageous for a design that allows multi-motor configurations — it can decrease dynamic efficiency. A large propeller increases aerodynamic torque on the propeller at steady state, causing the propeller to spin more slowly for the same input mechanical power. A slower propeller produces a decreased thrust, and therefore a decreased efficiency, at larger forward airspeeds since its slower rotational speed causes the propeller to “bite” into the air less. A propeller's dynamic thrust can be modeled as a linearly decreasing function of airspeed (Eq. 4.8):

$$T(V) = T(0)\left(1 - \frac{V}{p \cdot n}\right) \quad (4.8)$$

Where $T(0)$ is static thrust, n is propeller rotational speed in rotations per second and p is propeller pitch, which defines the AoA of a propeller blade. For a constant pitch propeller, the type used on model aircraft, a smaller diameter increases n and thus increases dynamic thrust and dynamic mechanical efficiency (propulsive power, or thrust times velocity, over mechanical power).

Thus, for a single-motor aircraft with a fixed-pitch propeller, a tradeoff in propeller diameter must be made between dynamic and static mechanical efficiency. However, in a multi-motor aircraft, multiple smaller propellers can match the total propeller disk area of a single larger motor while still achieving high rotational speeds and thus achieve good static and dynamic efficiency. Smaller propellers are additionally less limited by ground clearance; propeller analysis favors a multi-motor configuration.

Motor efficiency varies with steady-state rotational speed and K_v (or, equivalently, torque and the torque constant $1/K_v$) [13]. For a given battery voltage and propeller diameter, the propeller will reach some rotational speed, and thus K_v can be varied to achieve the same motor efficiency regardless of propeller diameter. Motor efficiency is therefore independent of propeller diameter, and is tuned by motor choice (performed in section 4.4.4).

One drawback of a multi-motor configuration is the additional weight of a second motor. This is important on the *Atlas* because on M2 thrust is limited by rated motor power as opposed to range. To rule out this possibility, the Avionics division conducted a survey of all brushless electric motors found on the popular website Hobbyking. The data reveals that motor mass correlates linearly with power rating (Fig. 4.12). Thus, two motors weigh about the same as one for the same total electrical power. WUDBF

therefore chose a two motor configuration to achieve both good static efficiency and dynamic efficiency for the same motor mass.

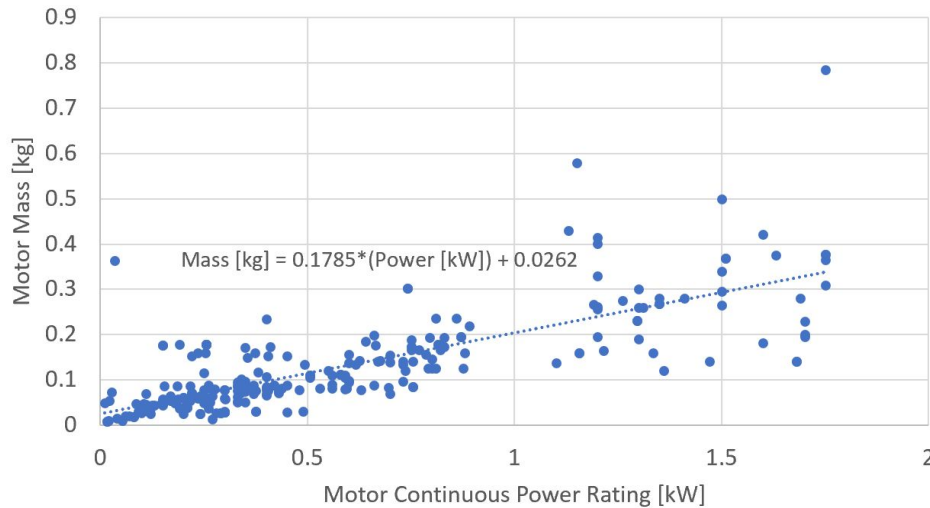


Figure 4.12: Plot of Hobbying brushless motor masses and power ratings. A trend line demonstrates the linear correlation between the two parameters.

4.4.3 Battery Specifications

The battery was chosen to maximize electrical efficiency and fit competition specifications. Since LiPo batteries have higher specific energy than nickel-metal hydride and nickel-cadmium cells, LiPos were chosen. To minimize electrical losses, the highest voltage possible was used for the LiPo pack: 6S. Power dissipated by resistance in the circuit is:

$$P_{diss} = I^2 R_{sys} \quad (4.9)$$

where I is the current and R_{sys} is the resistance of the system. This implies that for the same electrical power, a higher voltage dissipates less energy as heat and also reduces battery voltage sag

$V_{sag} = I \cdot R_{sys}$. Since competition rules require that LiPo batteries be placed in parallel, 6S batteries must comprise the pack. Few off-the-shelf batteries exist with higher than 6S voltage, which would severely limit flexibility in battery choice, so the 6S voltage was chosen. To stay within competition requirements batteries must, at most, hold 100 Wh each. Thus, to achieve the desired 200 Wh pack, 2 100 Wh batteries are needed in parallel. The 6S voltage constraint implies that these batteries must have a charge capacity of at most $100 \text{ Wh} / (6 \cdot V_{nom, LiPo}) = 4504 \text{ mAh}$. Batteries with 4500 mAh exist on the market. Therefore, WUDBF chose a 2P configuration of 4500 mAh 6S LiPo batteries. Of the battery packs found to meet these specifications, the only one from a trustworthy brand that exceeds the estimated 150 Wh/kg specific energy and with a high enough discharge rate is the Tattu 4500 mAh 6S, 25C battery.

A fuse was also required for this battery. The selected fuse — the Littelfuse MEGA 225A — is rated for the total battery pack continuous discharge rating of $2P \cdot 25C \cdot 4.5Ah = 225 \text{ A}$. The fuse also has a time constant long enough to account for the battery's burst current rating.

4.4.4 Motor Specifications

M2 and M3 call for different design criteria, requiring a tradeoff between a propulsion system optimization. M2 has no endurance constraint, so the highest cruise speed for M2 is obtained by maximizing propulsive power. Dynamic thrust on M2 is therefore limited by motor power rating, which is in turn constrained by motor mass.

The endurance constraint of M3 limits electrical power output. The mission length of 10 min combined with the 200 Wh battery pack requires an electrical power at cruise of less than 1200 W. To achieve the efficiency required for this mission requires larger propellers than for M2. Larger propellers also increase the static efficiency for this mission, which helps the aircraft take off in the required 20 ft.

Iterative simulations and Fowler flap analysis suggest that the *Atlas* could easily take off for M3 in 20 ft with a T/W above 2.5. Thus, a 4.5 kgf aircraft requires a total static thrust of 10 kgf. With the addition of a small factor of safety, the desired thrust for the *Atlas* is 12 kgf. A custom propulsion simulation, based on the first-order motor model used by the motor-propeller simulator QProp [13], was then run for a number of propeller diameters to determine the range of required motor voltage constants (K_v) to achieve 12 kgf at the 6S LiPo voltage and a good electrical efficiency (Fig. 4.13).

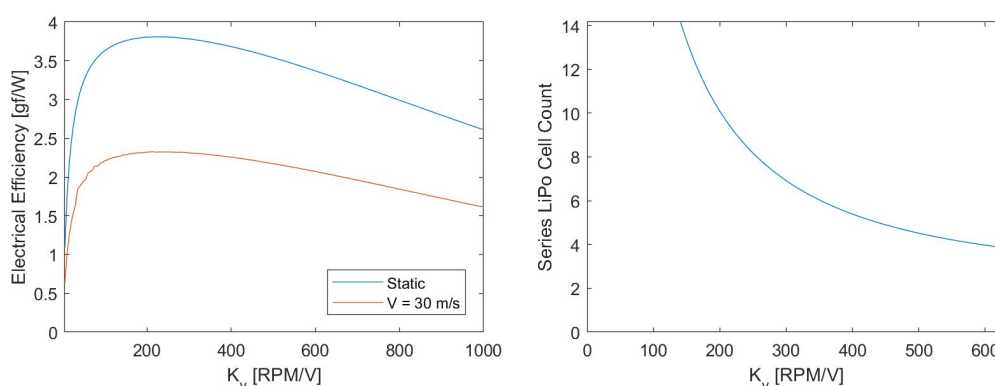


Figure 4.13: First order propulsion simulations for a 15 in. diameter and 10 in. pitch propeller on a single motor. Left: static and dynamic electrical efficiency over K_v with thrust held constant at 6 kgf. Right: required voltage, in units of LiPo cell nominal voltage, to achieve 6 kgf at different K_v .

Propeller diameter was varied until the K_v efficiency peak at left in Fig. 4.13 generally matched the K_v required with a 6S pack to achieve 6 kgf for each motor in Fig. 4.12. This ensures a good motor-propeller match at the steady-state rotation rate that overlaps the maximum efficiencies of both the propeller and motor. Pitch-diameter ratio (P/D) was assumed to be medium high ($\frac{2}{3}$) to achieve reasonable dynamic efficiency at larger diameters. This scheme achieved a good static electrical efficiency of over 3.6 gf/W for motor K_v ranging up to around 420. The simulation also predicted that for a 15 in. propeller diameter, a 350 K_v motor or higher would provide the necessary 6 kgf thrust per motor. This high static efficiency allows the plane to take off within the required distance. Figure 4.13 also shows

dynamic electrical efficiency at the estimated maximum speed of 30 m/s for M3. The plane will be flown at a lower throttle during cruise than on takeoff to meet the desired range, so for different estimated speeds this efficiency can be used to predict maximum average dynamic thrust.

Power output on M2, though, is limited by motor continuous power rating. For a total motor mass of less than 800 g, the trend line in Fig. 4.12 predicts a total power rating of around 4.3 kW. So the desired motor-propulsion system is around a 15 in. diameter, 10 in. pitch propeller on a ~400 g, 2.1 kW motor ranging from 350-420 K_v . Both a right- and left-handed (or E and EP) prop were desired so that the propulsion system produces no net torque on the aircraft, and in addition may decrease wingtip vortices.

The Sunnysky X4125 V3 420 K_v motor, with a mass of 351 g and continuous power rating of 2.25 kW, meets all of these requirements. Detailed time-domain simulations confirmed this motor-propeller design (Fig. 4.14).

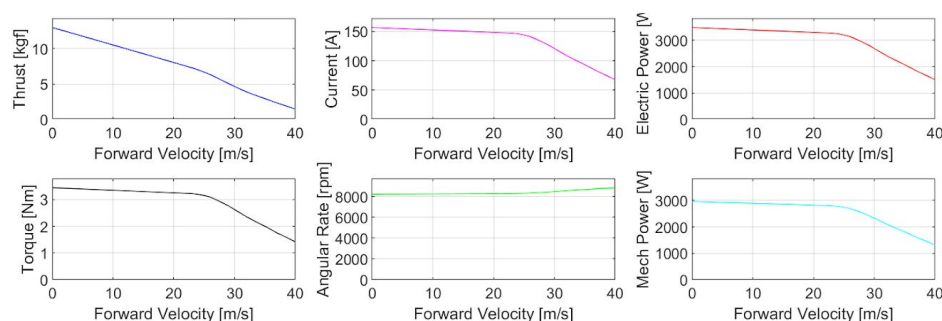


Figure 4.14 Detailed time-domain simulation of motor-propeller setup.

The propeller can change for M2. However, to increase thrust for the same battery and motor configuration, a larger propeller is required, or to increase dynamic thrust, a higher pitch is required. No market propellers exist with a diameter and pitch that would both increase thrust and stay within the motor and ESC rated currents and powers. Thus, the M2 flight will use the same propulsion package, but stay close to full throttle during cruise instead of reducing throttle for endurance like in M3.

The final propulsion package is given in Table 4.6. A summary of the propulsion system trade-offs is given in Table 4.7. A “+” implies advantages, a “-” disadvantages, and “=” means that the compared categories are similar.

Table 4.6: Propulsion package for all missions

Motor	Battery Pack	Propeller
Sunnysky X4125 V3 420 K_v	6S2P LiPo Tattu 4500 mAh 6S, 25C	APC 15x10, E & EP

Table 4.7: Propulsion design trade-offs.

Category	Trade-off Parameter and Score	
Motor Number	1	2
Motor Mass	=	=
Ground-limited propeller diameter	-	+
Complexity	+	-
Net torque on aircraft	-	+
Propeller Diameter	Large	Small
Two-motor static efficiency	=	=
Dynamic efficiency	-	+

4.5 Initial Performance Estimates

Tables 4.8 and 4.9 provide an overview performance of the *Atlas* as predicted by KM Sim.

Table 4.8: Estimated aircraft aerodynamic data by mission.

Mission Parameters	M1	M2	M3
CL Max (No Flaps)	1.19	1.19	1.19
CL Max (Flaps)	1.99	1.99	1.99
CD Cruise	0.091	0.094	0.097
L/D Cruise	3.23	3.95	4.59
Vcruise [m/s]	32	30	18.5
Vstall flaps deployed [m/s]	8.2	10.9	8.2
Vstall no flaps [m/s]	12.2	16.1	12.2
Takeoff Gross Weight [kg]	4	7.1	4.04

Table 4.9: Estimated aircraft performance by mission.

Mission Parameters	M1	M2	M3
Predicted Lap Time [s]	40	45.5	50
Number of Scoring Laps	3	3	12
Total Mission Time [s]	120	121	600
Number of Passengers	N/A	21	N/A
Score	1	1 + 0.72	2 + 0.63

M1 is projected to take 1.5 minutes, including takeoff. M2 is projected to take 121 s with lap times of 40.3 s and the *Atlas* carrying 21 passengers. After optimization and analysis, WUDBF increased the performance of the *Atlas* so that it can fly 12 laps in 10 minutes on M3 while towing the banner. KM Sim projects the *Atlas* to take off in 19.6 ft.

5. Detail Design

5.1 Dimensional Parameters

Table 5.1 contains details of the wing and control surface geometries, while Table 5.2 provides data regarding the weight, take off distance, and other hindrances on the *Atlas*'s flight capabilities.

Table 5.1: Wing and control surface geometry.

Wing Geometry Data	
Wingspan [in]	60.0
Root Chord Length [in]	13.5
Tip Chord Length [in]	4.91
Wing Area [in ²]	561
AR	7.16
Leading Edge Sweep [°]	22.6
Quarter Chord Sweep [°]	20.4
H-stab Length [in]	15.9
V-stab Length [in]	7.78
Fowler Flap Length [in]	43.9
Elevator Area [in ²]	24.5
Rudder Area [in ²]	10.9
Aileron Area [in ²]	41.4

Table 5.2: Restraints and geometry of the aircraft.

Aircraft Restraints/Geometry	
Aircraft Empty Weight [lb]	12.3
Aircraft Payload Weight [lb]	6.56
Aircraft Maximum Weight [lb]	18.9
Cruise Velocity [mph]	60
Available Static Thrust [kgf]	12.6
Available Dynamic Thrust at Cruise Velocity [kgf]	4.66
Take-off Distance [ft]	19.6

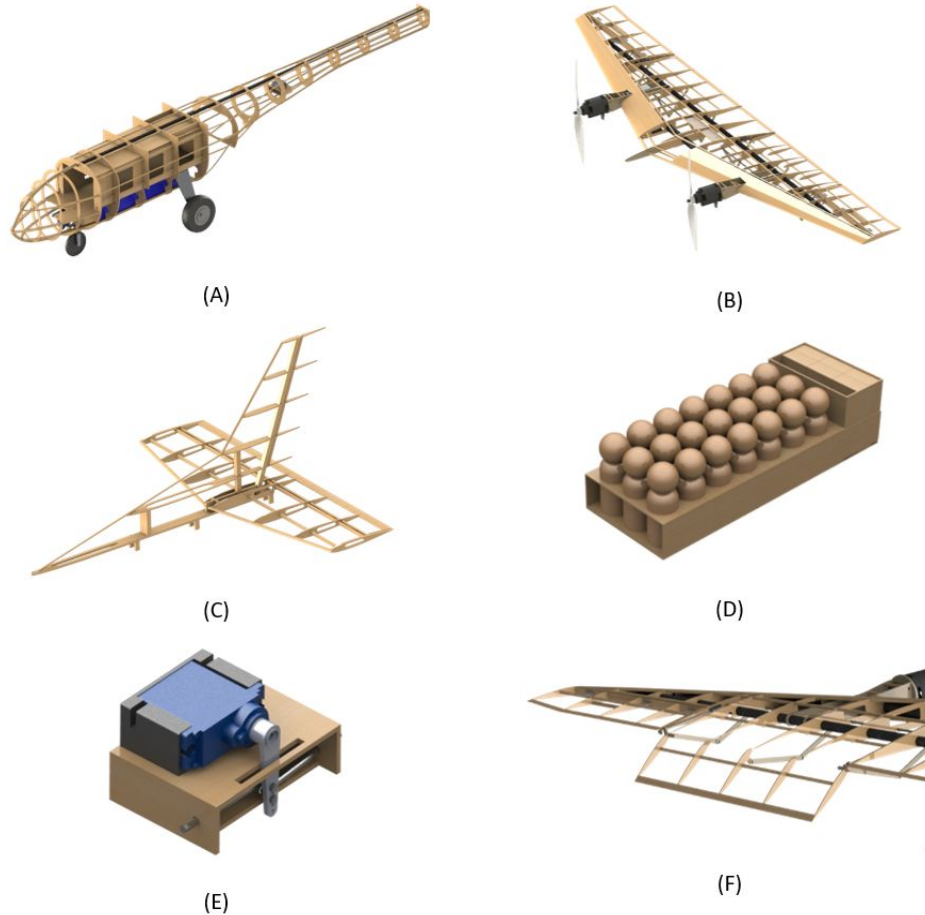
5.2 Structural Characteristics

The structure of the aircraft is composed of wood ribs connected by spars and stringers. The stringers are placed along the outer mold line of the airframe to increase the area moment of inertia of the wing and fuselage cross sections. To increase bending and torsional stiffness, carbon fiber (CF) spars are added to the wing and fuselage. Once completed, the structure is covered with UltraCote.

Spars are placed in the wing, fuselage, and empennage to ensure rigidity. Solidworks FEA was attempted for the placement of the spars. However, no computer was available that was capable of solving the complex geometry of the wing structure. As it is critical to keep the shear center as far forward as possible, the main spar in the wing is placed as far forward as possible while still extending through most of the wing span. To ensure both spars run uninterrupted through both sides of the wing, they are placed parallel to the Y direction rather than at the sweep angle. The secondary spar in the wing is placed further forward, allowing it to be thicker, but also not covering more than half of the span. Its placement allows the motor mounts to be placed at its ends. A single CF spar is also placed in the fuselage to ensure rigidity and provide a hard attachment point for the empennage and wing. In the empennage, wooden spars are placed at the thickest point in both. The combination chosen is recorded in Table 5.3.

Table 5.3: Spar placement and composition.

Part	Chord position (Center)	Material	Outer Diameter [in]	Inner Diameter [in]	Thickness [in] (square)
Wing	0.29	Carbon Fiber	0.715	0.625	n/a
Wing	0.58	Carbon Fiber	0.49	0.375	n/a
Fuselage	n/a	Carbon Fiber	0.34	0.25	n/a
HStab	0.4	Basswood	0.25	n/a	n/a
VStab	n/a	Balsa	n/a	n/a	0.25

5.3 System and Subsystem Design**Figure 5.1: CAD renderings of (A): Fuselage, (B): Wing, (C): Empennage, (D): Passenger compartment, (E): Banner mechanism, (F): Deployed Fowler flap.****5.3.1 Fuselage**

The fuselage stores electronic components, carries payloads, and connects the wing and tail. Bulkheads are made from Lite plywood. The nose cone is split in top and bottom halves. The upper half is connected to the fuselage by a hinge to allow access to the avionics and passenger compartments, while the lower half is rigidly connected to the front bulkhead. The largest section of the fuselage is a compartment designed to hold 21 passengers with luggage. Batteries are stored directly below, with 2 in. of horizontal clearance to adjust the CG. A CF spar runs lengthwise through the fuselage, increasing

stiffness and providing a mounting point for the tail. Initially, two more spars were placed lower in the fuselage to provide mounting points for the main landing gear, but were removed after the landing gear was redesigned following the first flight test. To save weight and create wire pathways, excess material is cut from the bulkheads and attachment plates. The fuselage is shown in Fig. 5.1A

5.3.2 Wing

The wing consists of 21 balsa ribs with two CF spars. The spars are positioned as far forward as possible to move the shear center in front of the aerodynamic center. To increase main wing leading edge stiffness and ensure that the curvature was correct between the ribs, the front quarter chord of the wing is wrapped with a $\frac{1}{16}$ in. balsa skin. The leading edge is a square balsa spar inset into the ribs and sanded down to match the curvature of the airfoil. Because of the sweep of the wing and decreasing thickness of the ribs, it is impossible to run the CF spars through the entire wing. The largest spar has a diameter of 0.715 in. and runs through seven ribs, while the smaller spar has a diameter of 0.49 in. and runs through seventeen ribs. To stiffen the outer section of the wing, a $\frac{1}{8}$ in. balsa beam runs through the outer five ribs on either side. This is oriented vertically to increase the area moment of inertia, and is a mount for the hinges of the aileron. Both motors are mounted directly next to the ends of the thickest spar. This gives a minimum distance from the center of the motor to the fuselage of 8.6 in., allowing for a maximum 16 in. diameter propeller. The motor mounts are made of Lite plywood, and the motor is placed in front of the leading edge of the wing while linking to both of the CF spars, distributing the load. The wing mount consists of two plates, one in the wing and one on the top of the fuselage to allow for removal. The plates are attached using two 10-32 bolts, which screw into T-nuts in the plate on the fuselage. The bolts are placed 4 in. apart in the Y-axis to maximize the reaction moment. A plywood beam extends past the leading edge of the wing in the middle, and slots into a bracket connected to the spar in the fuselage, providing three points of contact for the wing. The wing is shown in Fig. 5.1B

5.3.3 Empennage

The empennage is designed using the same rib construction as the main wing. Due to the smaller scale of both the vertical stabilizer and horizontal stabilizer, a $\frac{1}{4}$ in. diameter birch spar is added to the horizontal stabilizer for rigidity. A $\frac{1}{4}$ in. square balsa spar is placed in the vertical stabilizer for rigidity and mounting the rudder hinges. The birch spar is used to connect the horizontal stabilizer to a plate at the base of the vertical stabilizer, and the bottom rib of the vertical stabilizer is connected to the main spar of the fuselage. The attachment points to the bulkheads are each spaced so that the horizontal stabilizer is attached at a 1° incidence angle. The empennage is shown in Fig. 5.1C

5.3.4 Landing Gear

The initial design of the aircraft included retractable landing gear to reduce drag in flight. To avoid the challenges of manufacturing retractable landing gear, the actuators and struts were purchased from Robart Manufacturing. The actuators were rated for 12-18 lb aircraft, which is sufficient for the AUW of the *Atlas*. The nose gear strut is steerable using a servo. The main gear were originally attached to CF

square rods in the fuselage using birch plywood brackets. The nose gear actuator was connected to a birch plywood plate attached to the front bulkhead. To increase the stability of the gear while rolling, the nose gear was angled forward and the main landing gear was towed in.

Through testing, several inadequacies of the initial landing gear design were discovered (see section 8.5). The wooden brackets mounting the actuators to the CF rod proved very prone to failure, even when manufactured with birch plywood. Also, upon landing during the first flight test, the CF rod itself cracked along the axial direction, likely due to being a pultruded rod and therefore weaker in torsion. Consequently, it was decided that, due to time constraints and difficulties manufacturing the correct components, the landing gear would no longer be retractable. To reduce the weight and resolve the mounting issue, the next iteration of the main landing gear will be a solid-spring type while the nose gear remains unchanged. Two materials were considered for the strut, aluminum and carbon fiber. The strut was designed using sizing equations for solid spring landing gear, Euler-Bernoulli Beam theory, and estimating a maximum landing deceleration of 3g as the first test flight landed at 1g max [14]. The aluminum weighed 100 g with a factor of safety of 3. Using Solidworks Finite element analysis (FEA), the maximum von Mises stress was 258 MPa. With a yield stress of 505 MPa (7075T6 aluminum), the plate would have a final safety factor of approximately 2.

Despite the favorable results with the aluminum, an equivalent CF strut met the same sizing requirements while weighing less. Carbon fiber was selected over aluminum due to its lower weight, wider wheel span, and significantly lower manufacturing time.

5.3.5 Passenger Storage

The passengers are stored in a compartment toward the front of the fuselage and directly above the battery compartment. The compartment itself acts as a drawer, sliding partially out of the fuselage when the top half of the nose cone is removed, allowing the passengers to be loaded. Each of the passengers is placed in a slot in a “grid” such that the base of each passenger is restrained by the slot, and the middle of the passenger is restrained by another grid placed higher up. The passengers are then locked in place by rubber bands acting like seat belts. Luggage is stored behind the passengers in the same plane. Once the drawer is slid back into the compartment, the ceiling prevents any passenger from being removed from the grid. The passenger storage compartment is shown in Fig. 5.1D. This design enables an estimated loading time of 40 s.

5.3.6 Banner Release Mechanism

The banner is positioned behind the battery compartment on the bottom of the fuselage. The mechanism (Fig. 5.1E) consists of two components: the banner containment, and the release mechanism. The release mechanism was placed at the rear end of the banner, and both releases the banner initially and disconnects it from the plane at the end of the flight. The release mechanism uses two pins, each connected to the horn of a single servo. Two strings are connected to the pins, one holds the banner in place during flight while the other holds the banner when deployed. When the servo rotates one direction,

the first pin is released from the plate, releasing the banner from the plane. When the servo rotates the other direction, the second pin is released, detaching the banner entirely. The banner containment is a single plate that is connected to the bulkhead in front of the banner while stowed. The front end of the banner is secured by the plate, and restrained on the other side by the string connected to the release mechanism. Estimated loading time is 20 s.

5.3.7 Flap Mechanism

This design uses Fowler flaps, which must achieve the correct geometry to maximize lift while rigid enough to handle the maximum motor thrust. The flaps are made from balsa ribs with basswood spars and balsa plates to increase the rigidity enough to be covered with UltraCoat without warping. The flap is attached to a slide on either side. These are attached to the ribs on either side of the flap. This allows the correct amount of translational motion in the Fowler flap. To achieve the rotational motion three hinges are attached, one to either side and one in the center. The hinges are made from two Lite plywood beams of varying lengths. The middle hinge has a slightly shorter beam, causing it to “pull” on the flap as it extends, rotating it to 39° deflection. The hinges are rigidly attached to the same $\frac{1}{8}$ in. steel rod rotated by a servo. The lengths of the beams required to achieve the correct geometry were verified using a SolidWorks motion study. The flap mechanism is shown in Fig. 5.1F.

5.3.8 Electrical Diagram

Figure 5.2, an electrical schematic of the whole aircraft, clarifies the arrangement and interaction of the various electrical subsystems.

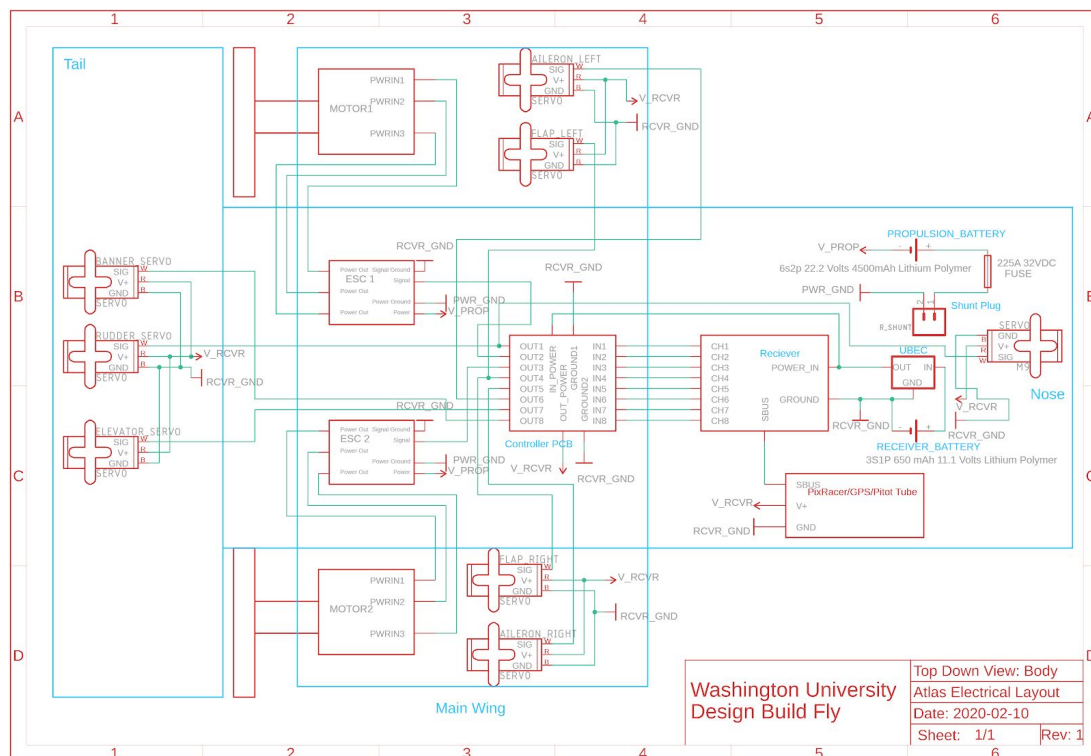


Figure 5.2: Electrical layout of the Atlas.

5.3.9 Servo Selection

To select the servos on the aircraft, the torques on them must be estimated. To make this estimation, the Aerodynamics Division performed maximum aerodynamic hinge moment predictions from XFLR5 using a fast, inviscid method. A MATLAB script was then written to predict moments on a servo horn given the control surface hinge moment and the system geometry: the radius of the control horn, the radius of the servo horn, the horizontal distance between servo axis and control axis, and the vertical distance between servo axis and control axis. The torque was calculated up to a maximum control surface deflection of 30° , shown in Fig. 5.3.

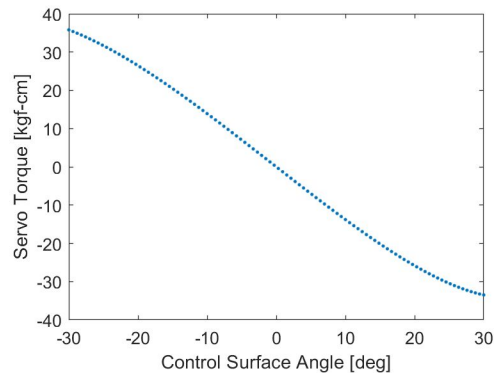


Figure 5.3: Graph of typical servo torque over control surface angle.

With the maximum moment on the servo predicted, a factor of safety for servo torque was estimated. XFLR5 cannot calculate fully viscous flow for a wing with deflected surfaces; as a result, hinge moment caused by drag is neglected. This uncertainty was estimated to require a factor of safety of 2. Another factor of safety of 2 was applied to account for decreases in servo response time due to torque and from friction, dynamics, servo quality, and other effects. After the total factor of safety of 4 was applied to the torques, servos were chosen to minimize weight. Chosen servos for each mechanism are outlined in Table 5.4. All servos chosen were also used on last year's vehicle, which saved the time required to test each servo.

Table 5.4: Chosen servos, torques, and quantity on each mechanism.

Part	Servo	Quantity
Ailerons	BMS-A53M HV	2
Rudder	DS-929HV	1
Elevator	BMS-101HV	1
Flaps	DS-929HV	2
Nosewheel	DS-929HV	1
Banner	BMS-101HV	1

5.3.10 ESC Selection

The electronic speed controller (ESC) was sized for the predicted power requirements of the propulsion system and to minimize mass. The T-motor 80A T80A ESC exceeds the required 80 A/motor specifications (Fig. 4.14) and has a mass of 62 g, lower than or similar to the masses of all other ESCs

surveyed. This ESC saves 30g off of the 100 A Castle Edge Lite chosen for the previous iteration of the propulsion package, which had a mass of 92 g and required recalibration for every flight (see section 8.1).

5.3.11 Receiver Battery Selection

The aircraft uses various servos to control the plane and the banner release mechanism. Receiver pack voltage is constrained by servo voltage requirements: a constant 7.4 V provided by a Universal Battery Eliminator Circuit (UBEC) satisfies all servos. The UBEC, a HobbyWing UBEC 10A (2-6S), was chosen to satisfy the servo voltage and current requirements (Table 5.5) and has a greater than 90% efficiency. It needs an input voltage of 8.6 V to output 7.4 V at all battery charge states, so the receiver battery must be at least a 3S LiPo pack (11.1 V).

Servo currents were calculated by assuming all control surface servos are simultaneously stalled for the longest flight of 10 minutes and other servos are at their rest currents for the flight (Table 5.5). This current estimation provides an energy buffer as it is highly unlikely for the aircraft to operate with all flight critical servos stalled for the whole flight. Servo currents were measured for last year's aircraft. The resulting desired charge capacity of the 3S receiver pack is multiplied by 2 as a safety factor. This safety factor accounts for cold temperatures, which decrease battery voltage, and pre-flight loitering of the aircraft. The desired charge capacity is thus:

$$(2.02 \text{ A}) \frac{10 \text{ min}}{60 \text{ min/hr}} (7.4 \text{ V}) \frac{1}{0.9} \frac{1}{11.1 \text{ V}} (FoS) = 0.25 \text{ Ah} \cdot FoS = 0.5 \text{ Ah} \quad (5.1)$$

The team therefore decided to use a Tattu 800 mAh 7.4 V 45C 2S1P Lipo battery back, which slightly exceeds requirements, but weighs 42 g. No lighter packs between 500 and 800 mAh were found.

Table 5.5: Servo currents, including per-servo and overall currents.

Servo Name	# of Servos	Predicted Current [A]	Rest Current / Servo [A] [9]	Stall Current / Servo [A] [9]
BMS-A53M	2	0.8	0.02	0.4
DS-929HV	4	1.1	0.01	0.4
BMS-101HV	1	0.01	0.01	0.3
Total on Plane	7	2.0		

5.3.12 Microcontroller Development

The main purpose of using a microcontroller is to record flight data during test flights to iteratively improve the design. A Pixracer microcomputer was chosen to perform this task because the computer can run advanced flight control software like the open source PX4 software stack, and has the option of adding custom code on top to customize the system for the aircraft dynamics. All these features, while not necessarily expected to be used during competition, are useful for testing and analysis. This system utilizes the SBUS protocol to communicate serially with the receiver. Onboard sensors include a GPS, three magnetometers to sense compass orientation, a barometer, a six-axis inertial measurement unit (IMU), a pitot tube, and a voltage and current sensor. All are temperature compensated. All flight data is recorded on a micro SD card and transferred to a computer for post-flight analysis.

To reduce wiring complexity and account for the different power requirements of the servos and the microcontroller, a Printed Circuit Board (PCB) was designed with eight input ports and eight output ports that take the control inputs from the receiver and send them to the servos using power from the battery elimination circuit. The servos draw power directly from the receiver pack instead of pulling all of their current through the receiver itself, protecting the receiver from over-currents. These eight outputs are then connected to multiple servos, some (such as the nosewheel and rudder) through Y-harnesses.

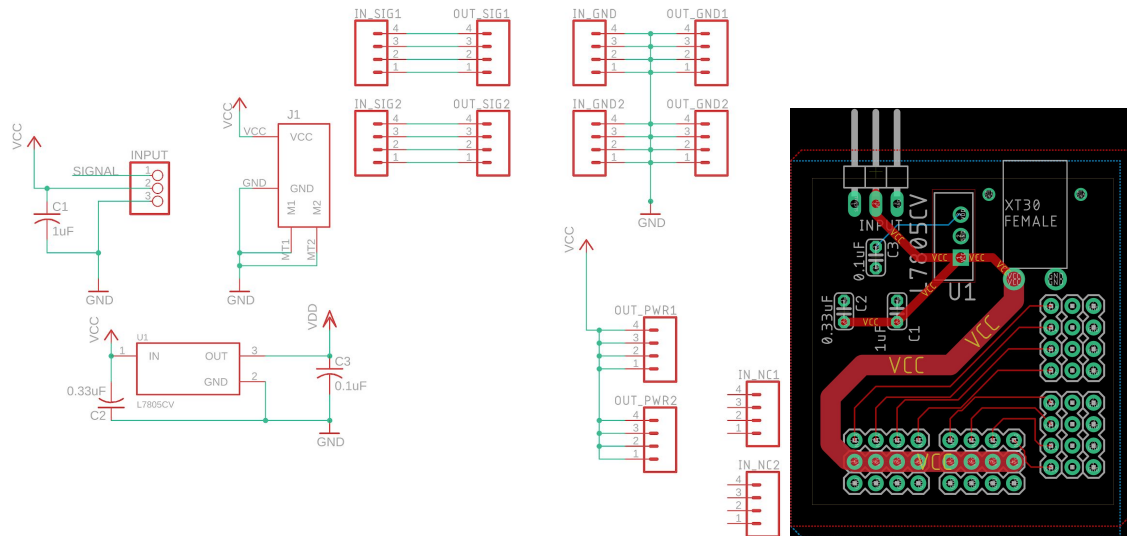


Figure 5.4: Left: schematic for the servo control board). Right: PCB layout (red traces are on top, blue are on bottom); ground planes not shown for clarity.

5.4 Mass and Balance

Each mission has a different mass distribution. To estimate the CG of the vehicle, component masses and locations were determined from the CAD model and tabulated. Once found, the components may be referenced using absolute coordinates in the traditional body axis. The tip of the nose, was taken to be the origin, and a mutually orthogonal coordinate system was established, with the positive X-axis pointing out from the nose of the aircraft, positive Z-axis extending below the aircraft, and the positive Y-axis directed out of the right wing, relative to the nose facing forward. The masses and absolute coordinates of the aircraft's major components are listed in Table 5.6. Note that components such as bolts, glue, covering, and wire mass are not included in the table.

Table 5.6: Mass and CG table.

Mission	Mass [kg]	X [in]	Z [in]
Wing Structure	.35	-21.8	-5.68
Fuselage Structure	.19	-22.2	-2.19
VStab Structure	.021	-56.1	-6.11
HStab Structure	.015	-57.2	-4.75
Landing Gear (Main)	0.27	-21.6	0.67
Landing Gear (Nose)	0.19	-8.46	1.46
Electronic Components	0.64	-15.8	-1.4
Main Battery	1.5	-17.5	1.56
Motors	0.83	-14.9	-5.18
M1 (empty)	4.06	-17.5	-1.26
Passenger Compartment (Empty)	0.033	-17.1	-0.74
Banner Mechanism (Empty)	0.023	-37.2	-3.42
M2	7.06	-16.9	-1.47
Passenger Compartment (Full)	3	-16	-1.75
M3	4.10	-17.5	-1.26
Banner Mechanism (Full)	.043	-37.2	-3.42

5.5 Flight and Mission Performance

Table 5.7 provides updated aerodynamic properties and Table 5.8 provides performance data synthesized from flight tests and KM Sim updated with experimental wind tunnel and thrust data. Data was normalized using other aircraft created in section 3.4. Ground Mission score was estimated based on best guess layouts and times for other aircraft and other teams' times.

Table 5.7: Parameters for the *Atlas* KM Sim.

Flight Parameters	M1	M2	M3
CL Max (No Flaps)	1.2	1.2	1.2
CL Max (Flaps Deployed)	3.2	3.2	3.2
CD Cruise	0.05	0.05	0.06
L/D Cruise	9.4	5.6	12.6
Vcruise [m/s]	21	39	15
Vstall flaps deployed [m/s]	9.1	10.7	9.1
Vstall no flaps [m/s]	13.6	16.5	13.6
Takeoff Gross Weight [kg]	4.9	7.9	4.94

Table 5.8: The *Atlas* KM Sim performance.

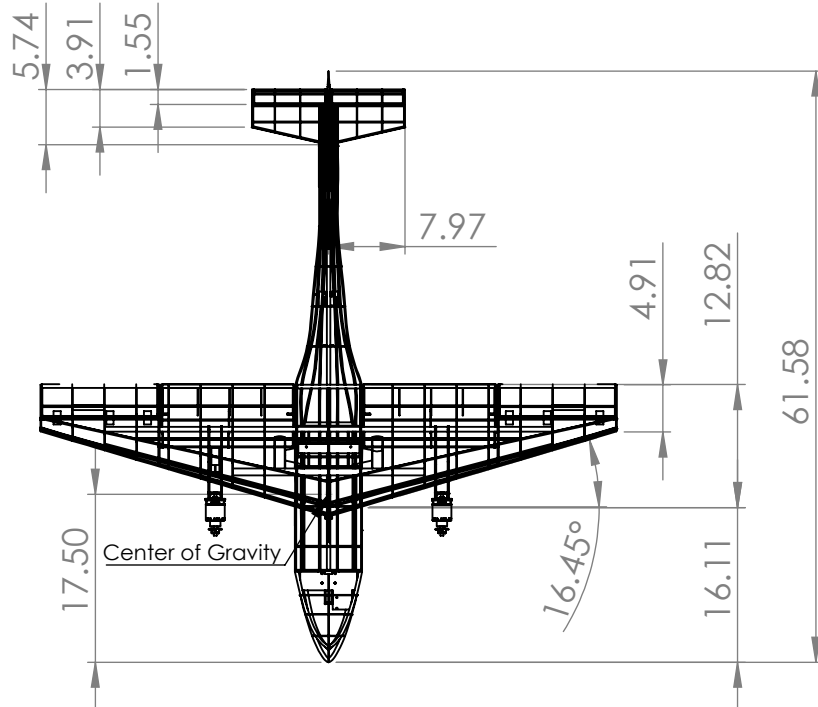
Mission Parameters	M1	M2	M3	GM
Predicted Lap Time [s]	43.3	39.7	59.4	n/a
Number of Scoring Laps	3	3	10	n/a
Total Mission Time [s]	130	119	600	60
Number of Passengers	n/a	21	n/a	21
Score	1	1 + 0.724	2 + 0.6	0.5

5.6 Drawing Package

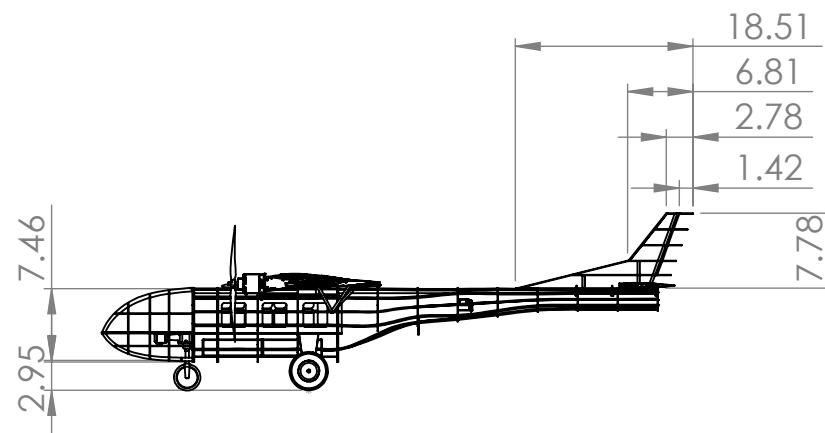
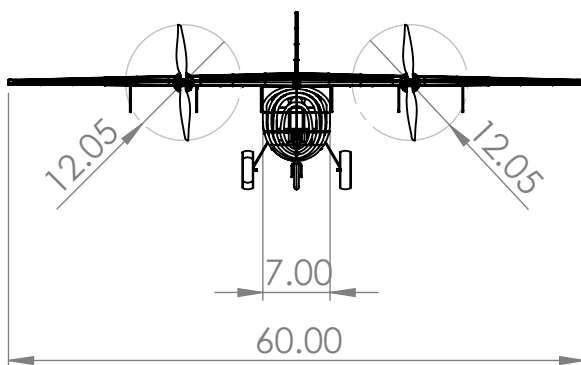
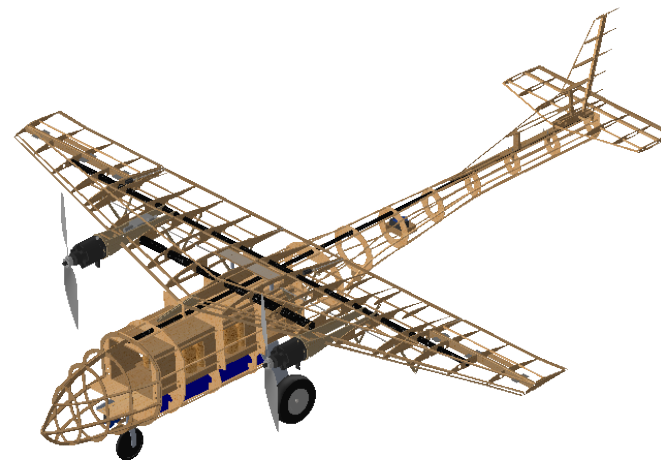
Detailed in the following pages is the drawing for the *Atlas*. Drawings are, in order, three-view, structural layout, systems overview, and layout of payload accommodations.

2

1



Scale 1:15



All dimensions are in inches

COMMENTS:

DRAWN BY:

REVIEWED BY:

Washington University in St. Louis
AAIA Design/Build/Fly 2020

TITLE:

Atlas

DWG. NO.

Aircraft 3-view

SCALE: A:A

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SHEET 1 OF 1

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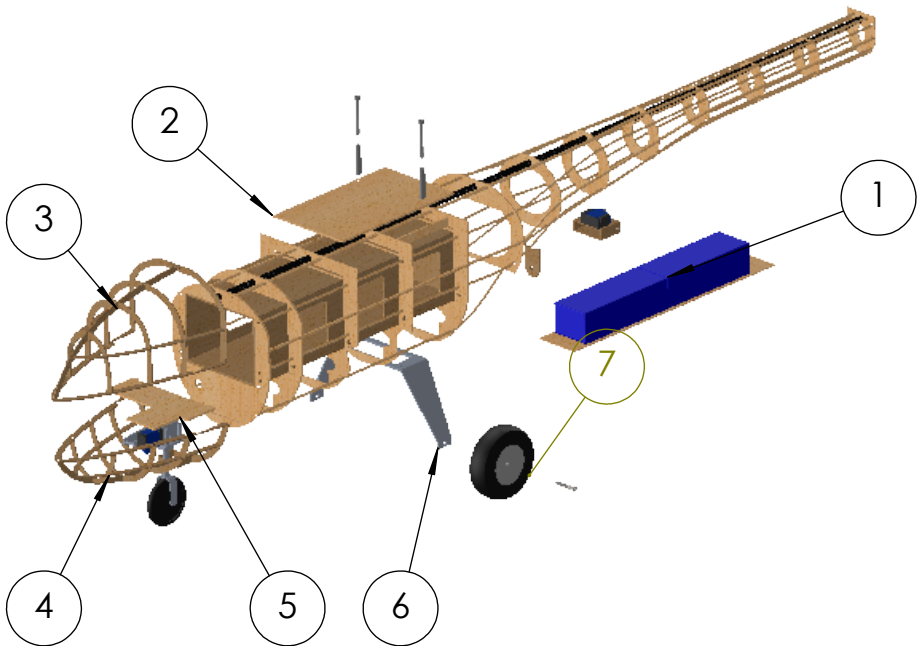
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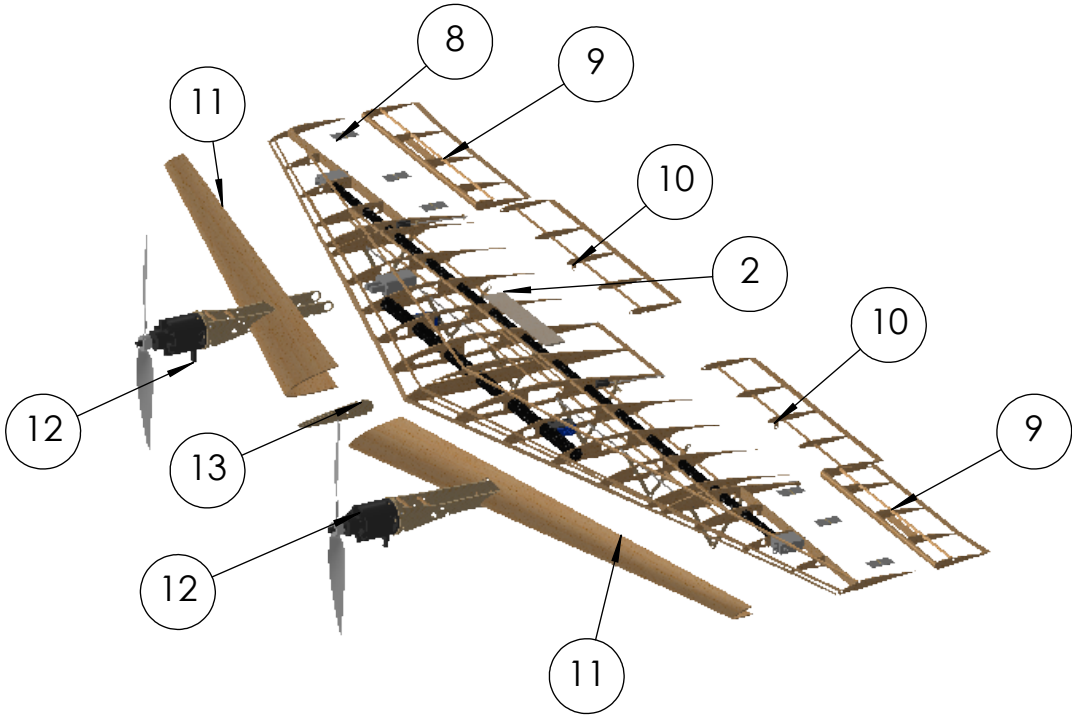
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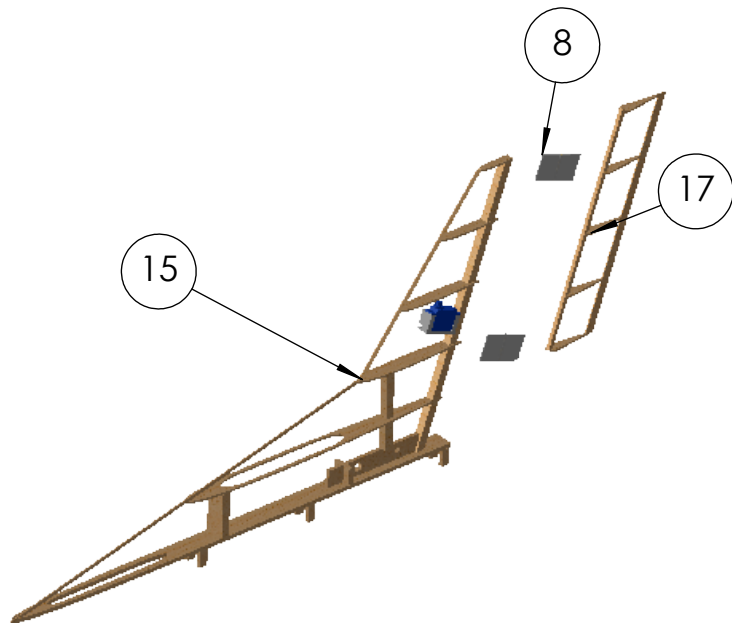
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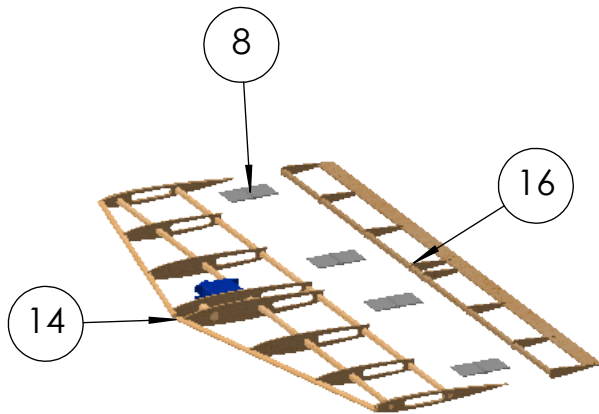
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Atlas

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Structural Layout

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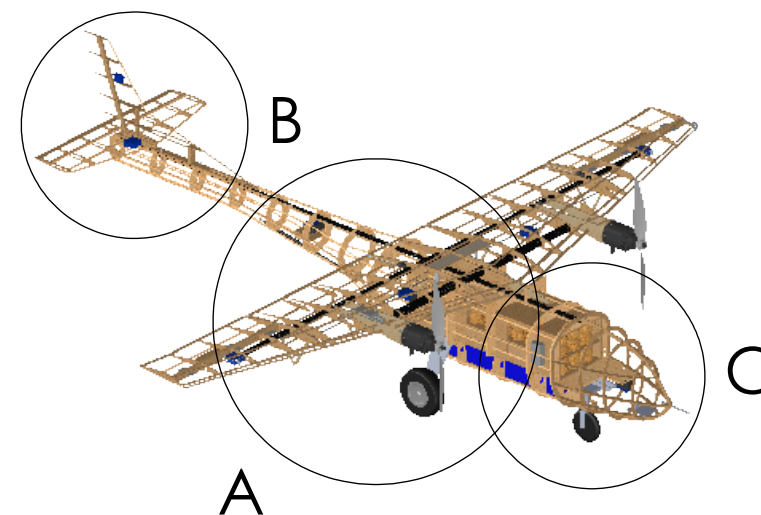
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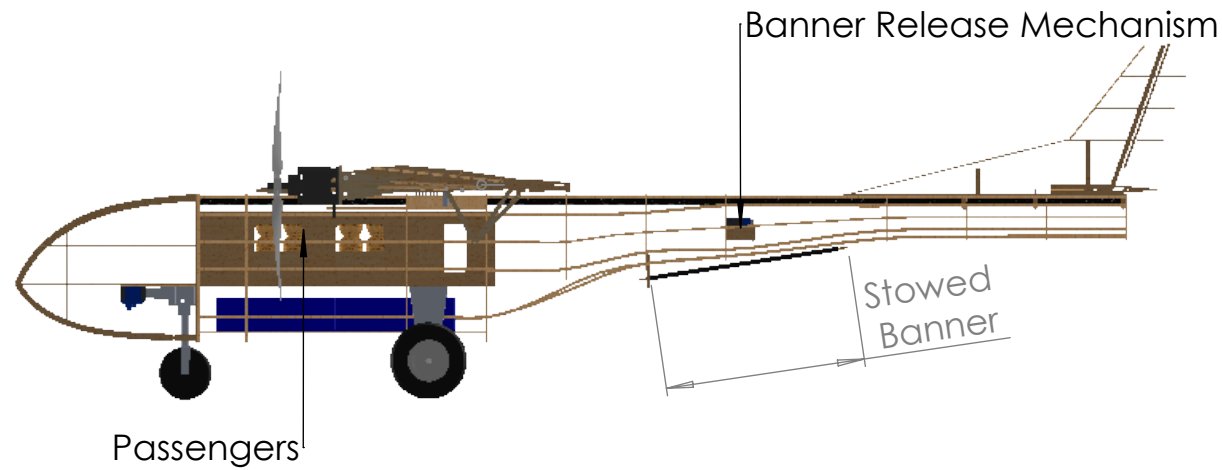
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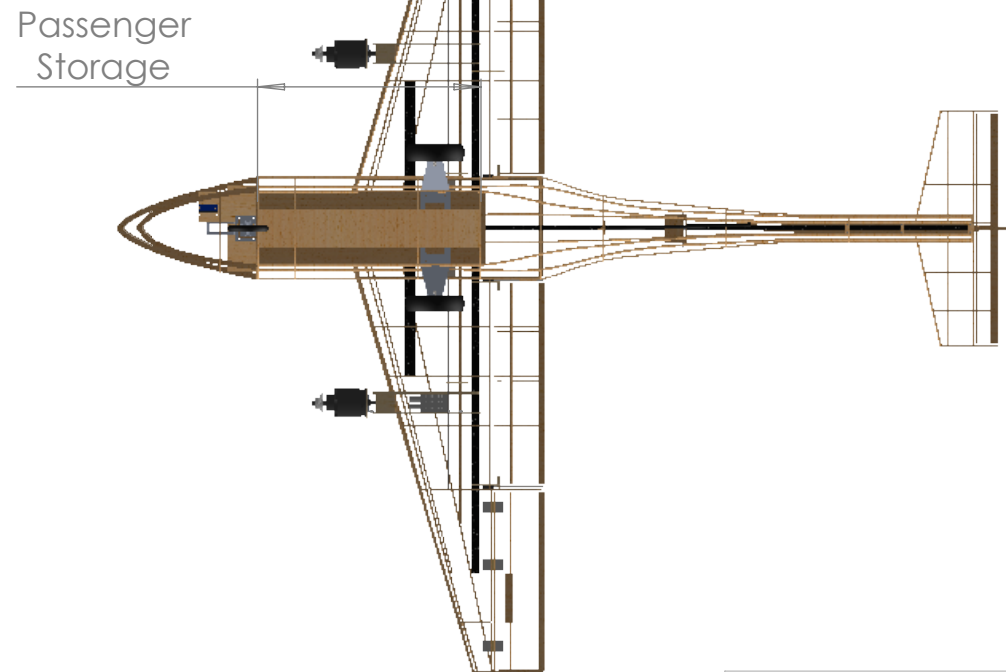
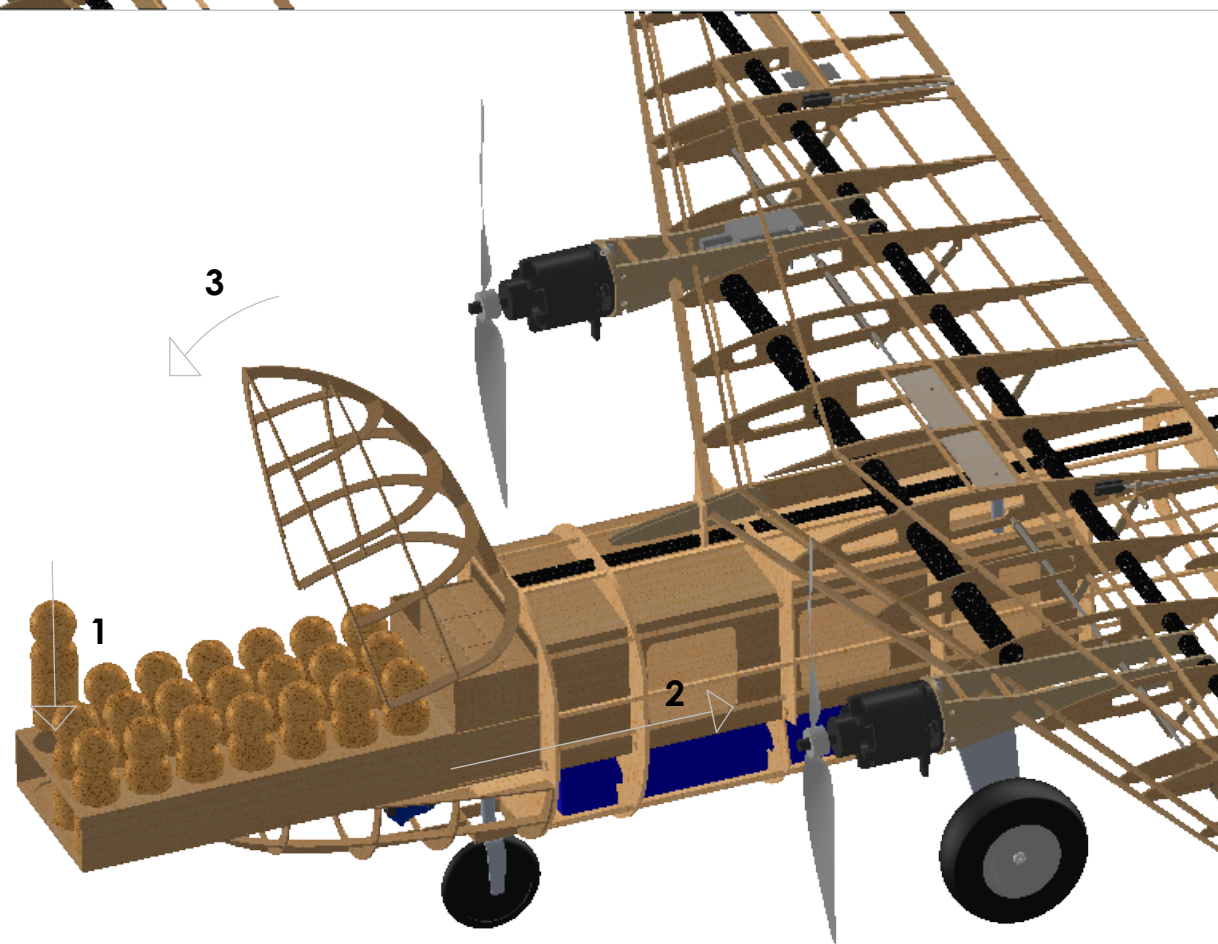
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COMMENTS:

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Washington University in St. Louis
AAIA Design/Build/Fly 2020

TITLE:	DWG. NO.	
Atlas	Payload Accomadation	
SCALE: A:A	WEIGHT:	SHEET 1 OF 1

6. Manufacturing Plan

6.1 Manufacturing Methods Considered on Major Components

6.1.1 Selection Criteria

Decreasing manufacturing risk played a significant role in the development of this year's aircraft. As such, the Manufacturing Division sought a manufacturing method which enables rapid prototyping while still flexible and forgiving. Thus the importance of Ease of Iteration was set to 0.25. Cost of prototyping was also taken into consideration, with a weight of 0.10. For issues not immediately caught, flexible construction methods would allow for adjustments, maintenance, and repairs. Flexibility was thus determined to be 0.20. These goals were considered along with expected strength, weight, reliability, and team experience, with importances 0.15, 0.10, 0.15, and 0.05, respectively.

6.1.2 Solid Foam

The Ease of Iteration was scored at 2. Damaged foam construction cannot be repaired and must be replaced, meaning a Flexibility score of 2. Finally, a fuselage with a small cross section cannot be structurally supported by foam: Expected Strength and Reliability were scored at 4. Expected Weight and Cost scores were estimated as 5 and 4, respectively. As WUDBF built a foam plane last year, it received an Experience score of 8.

6.1.3 Additive Manufacturing

The availability of 3D printers makes it appealing. Its rapid prototyping ability warranted an iteration score of 7 and a Flexibility score of 4. However, 3D-printed structural components have historically failed. The printing process was also unreliable and led to delays. Components were heavy, warp-prone, and split between print layers. It therefore received strength, reliability, and weight scores of 0. Past experience led to an Experience score of 8. Expected Cost was scored at 5. Additive manufacturing was thus only considered for small, nonstructural components.

6.1.4 Monocoque Methods

Fiberglass or CF monocoque methods were considered but ultimately dismissed due to their high costs, difficulty of maintenance, lack of team experience, and an inability to iterate quickly. The method scored low in these categories, as shown in Table 6.1. However, prefabricated CF rods and fiberglass as reinforcement were pursued further. Tests show that they could strengthen the wing and fuselage. As such, strength, weight, and reliability scored at 8, 5, and 5, respectively. The wing proved to be strong enough without fiberglass, so it was not needed.

6.1.5 Wood Construction

Wood construction was previously researched. It proved to be relatively cheap, light, easy to work with, and readily available. Alterations and repairs could be made easily. Access to a laser cutter enabled rapid prototyping. These factors resulted in high scores in all categories except experience. Exact scores in each category are shown in Table 6.1.

6.1.6 Methods Chosen for Major Components

A summary of the manufacturing methods is shown in Table 6.1.

Table 6.1: Score table for manufacturing methods.

Feature	Importance	Method			
		Solid Foam	Additive Manufacturing	Monocoque	Wood
Ease of Iteration	0.25	2	7	3	8
Flexibility	0.20	2	4	2	8
Expected Strength	0.15	4	0	8	7
Expected Reliability	0.15	4	0	5	8
Expected Weight	0.10	5	0	5	5
Expected Cost	0.10	4	5	2	7
Experience	0.05	8	8	0	4
Weighted Total Score	1	3.4	3.45	4.2	7.25

Wood construction was chosen as the primary manufacturing method for all major components (Table 6.1) due to its high score. The shape of the wing and fuselage were defined by laser cut ribs. Wood stringers ran the length of each. Both feature CF rods to increase rigidity. The tail assembly and control surfaces were made with balsa ribs and stringers. Landing gear was purchased from a third party and attached with birch ply mounts.

6.2 Manufacturing Process

6.2.1 Wing

The basic elements of the wing are the ribs and stringers. Ribs must be positioned correctly before stringers are installed: U-shaped jigs secure them during construction. The central 13 ribs were installed to allow the load bearing CF rods to be inserted. The motor mount pieces attach to the CF rods and so were also inserted. The remaining ribs were then similarly installed. Basswood stringers were added for structure and balsa stringers for the leading edge, where it was sanded to shape. Thin wing sections above the Fowler flaps were reinforced with balsa sheets, shown at right in Fig 6.1.

After the frame was completed, servos and control surfaces were installed, the jigs were cut off and sanded down, and the wing was flipped over. Remaining structural basswood sticks were added and the wing attachment plate was inserted. The motors were integrated at this stage. The leading edge was formed by wrapping balsa sheets soaked in warm water around a stringer, as determined in section 8.3.1. Sharper portions of the leading edge used two smaller sheets of balsa. A smoother UltraCote surface was achieved with additional balsa sticks. The complete wing is shown at left in Fig. 6.1.

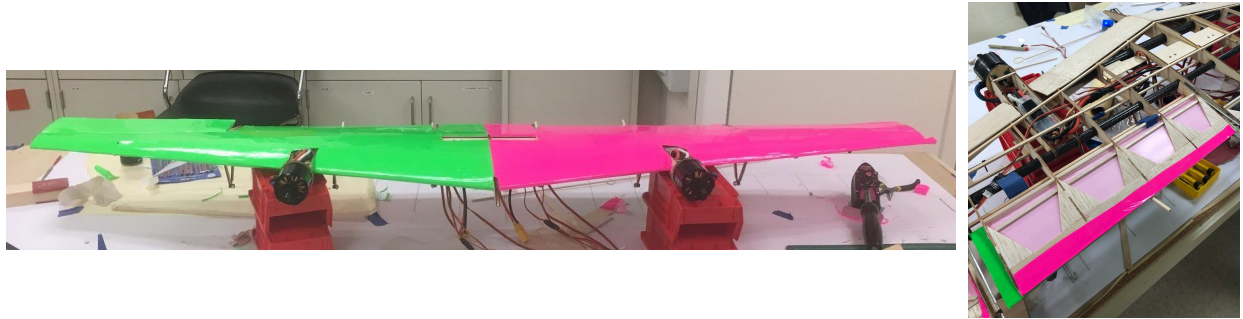


Figure 6.1: Left: complete main wing. Right: Fowler flap area trailing edge reinforcements.

6.2.2 Control Surfaces and Attachments

The *Atlas* has two ailerons, two Fowler flaps, a rudder, and an elevator. The main body of each was constructed using jigs for proper alignment. The frames for each control surface feature lateral basswood stringers. Trailing edges feature 1/16 in. balsa sheets sanded to half width. The ailerons also feature a sanded leading edge which matches the airfoil. Holes were cut into the leading edges (except on the Fowler flaps) to house plastic hinges which enable free movement of the control surface. Strength testing showed additional structure was required, accomplished with balsa sheeting and cross bracing.

Figure 6.2 shows the completed Fowler flap. Each Fowler flap had three arms linking the flap to a steel dowel. The outside arms connected to a pin which slides along a rod as the Fowler flap extends. The arms segments were connected by a screw and nut. The team added wooden attachments to the arms to keep the Fowler flaps secure while stowed. UltraCote was applied before the control surfaces were installed, shown at right in Fig. 6.2.

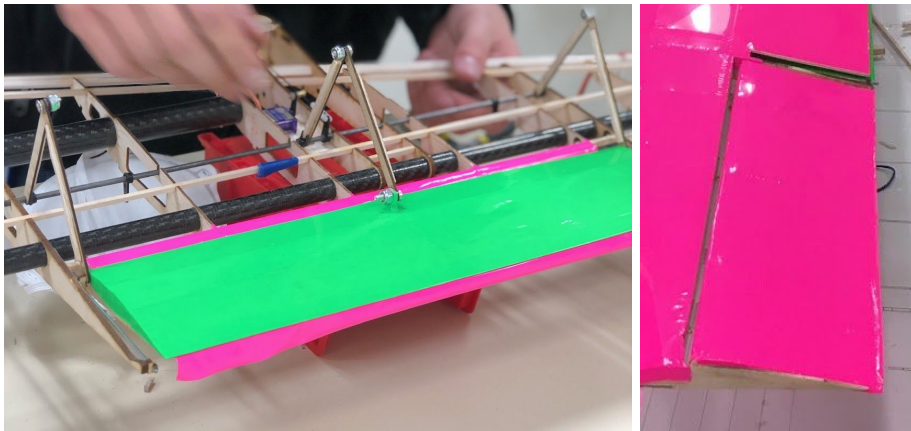


Figure 6.2: Left: coated Fowler flap. Right: coated aileron.

6.2.3 Fuselage

The fuselage is constructed from birch ply ribs and basswood stringers. Birch ply was chosen for its strength, and basswood was chosen for its higher relative strength and adequate flexibility—the fuselage design features curved stringers. A jig ensured proper rib alignment. Stringers were added from the tail forward to connect ribs. Some stringers were spliced for additional length. Once in place, the team

integrated the CF rods described in section 5.3.1. Next, platforms for the electronics and passengers were installed, shown in Fig 6.3. Finally, the fuselage was streamlined with UltraCote; however, this was done last in the manufacturing process to simplify systems integration.

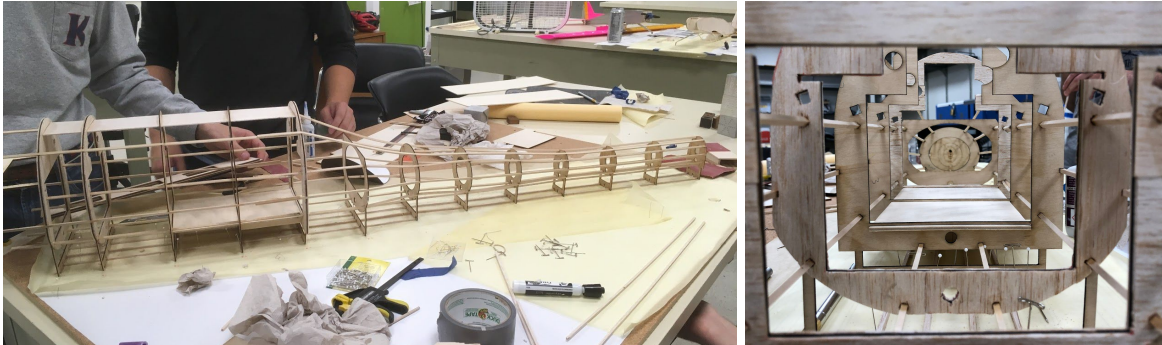


Figure 6.3: Left: fuselage construction. Right: bare fuselage interior.

6.2.4 Tail

The tail was constructed using basswood stringers, balsa wood, wooden dowels, and birch ply. Birch ply was used as the connection point to the fuselage. Basswood stringers served as the leading edges, for structural strength, and the trailing edges, for attachment points for control surfaces. These stringers were sanded to the shape of the desired airfoils. Basswood ribs are secured by wooden dowels, which also increases rigidity. The vertical and horizontal stabilizers were constructed separately, then joined to form the tail. Following electronic integration, the tail was covered in UltraCote. The tail construction is shown in Fig 6.4.

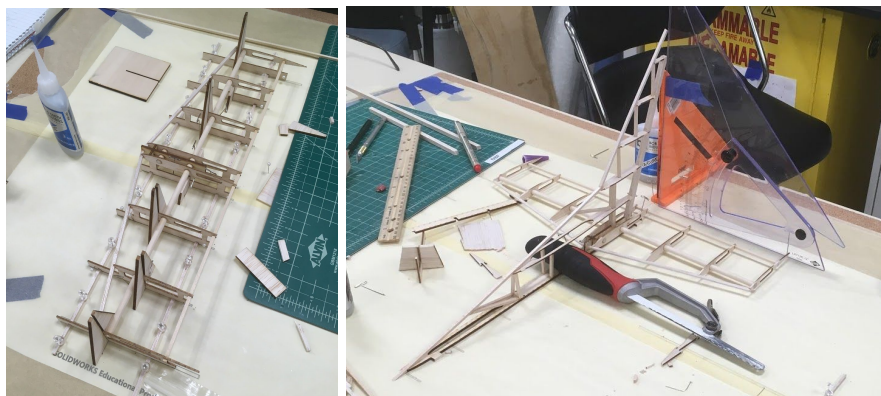


Figure 6.4: Left: horizontal stabilizer construction. Right: horizontal and vertical stabilizers.

6.2.5 Banner Mechanism

The banner release mechanism, shown at left in Fig. 6.5, was constructed using several laser cut balsa parts. These were connected using CA, and the release servo was attached to the balsa frame using a 3D printed mount. Each of the pins for the mechanism were made by cutting control rods short and adding a z-bend to attach them to the servo horn, which was extended using a Lite ply.



Figure 6.5: Left: banner release mechanism. Right: prototype passenger compartment.

6.2.6 Passenger Compartment

The passenger compartment was constructed from balsa wood to minimize weight. Passengers are secured by a balsa lattice with rubber bands around the passengers. The rubber bands are placed every row and hold three passengers together. The rubber bands are lightweight but strong enough to ensure passengers do not jostle or fall out during flight. The overall shape of the passenger compartment is rectangular and slides into the fuselage through a door below the tail. A prototype is shown in Fig. 6.5.

6.3 Manufacturing Milestones

The manufacturing schedule was implemented to ensure components were completed and tested on time. The prototype and the final aircraft were divided into important subtasks, as shown in Fig. 6.6.

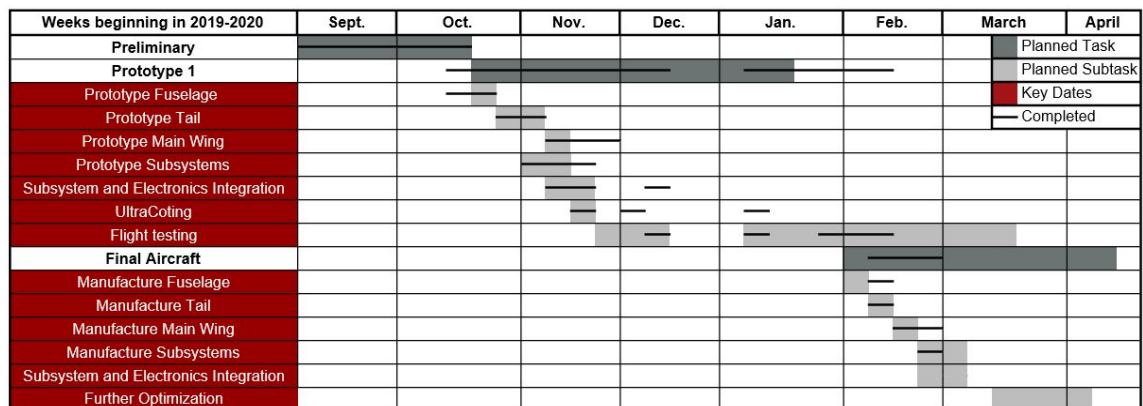


Figure 6.6: Gantt chart of the manufacturing plan, showing actual, projected and planned schedule.

7. Testing Plan

The WUDBF testing schedule is given in Fig. 7.1.

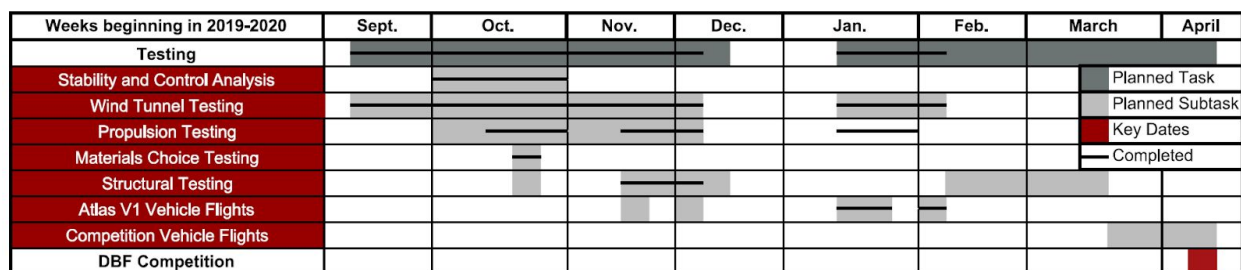


Figure 7.1: WUDBF testing schedule.

7.1 Propulsion Testing

The objective of propulsion testing is to verify the propulsion package in Table 4.7 matches simulated results, and to extrapolate better models from tested data. Propulsion testing is performed with an RCbenchmark Series 1585 dynamometer. Tests conducted include static thrust vs. throttle input to map throttle correctly in simulation and predict dynamic thrust during cruise, motor constant testing to determine empirical motor constants, and a full throttle static test to match simulation results. This testing allows simulations to better predict aircraft qualities and enables accurate trade studies. Detailed propulsion testing used an earlier version of the propulsion package. Rule changes to LiPo battery configuration forced a propulsion redesign, which requires additional testing to match simulations.

7.2 System and Subsystem Testing

7.2.1 Transmitter Testing

Before each flight, the transmitter is turned off to ensure the aircraft enters fail-safe mode in the case of a transmission failure and to meet the safety requirements. It ensures that the throttle is closed, the elevator is in the full up position, the right rudder and aileron are in the full right position, and the flaps are fully deployed. The transmitter is also tested for acceptable range (5,000 ft) whenever the receiver antennas are moved. The test also confirms the transmitter antennas are oriented orthogonal to each other, they are both parallel to the ground, and the dead zone of one antenna is not pointing toward the aircraft. The receiver antenna is also secured and unobstructed.

7.2.2 Banner Mechanism

To test the functionality of the banner release mechanism, a string was attached to one pin and a force applied at the end of the string to simulate the weight/drag of the banner. The BMS-101HV and DS-929HV servos were then actuated using a servo tester to release the string.

7.2.3 Fowler Flaps

As the most complex moving part on the aircraft, the Fowler flap mechanism was tested extensively. WUDBF verified the flaps would consistently deploy in full under full aerodynamic loading. To test full deployment, the flaps were extended and retracted with a servo tester. The team observed the behavior of the control rod, elbows, and pins. Successful deployment involves the rod turning without significant torsion and the elbows moving in unison. Pins must travel the length of the guide rails. The flaps must not droop when retracted and achieve proper deflection when deployed. Consistent deployment testing involves several deployment cycles. The flaps must successfully deploy several times in a row. To test deployment at full throttle, a secured aircraft must achieve the previous two testing cases while the motors run.

7.2.4 Control Surfaces

A similar testing system was used for all remaining control surfaces: ailerons, elevator, and rudder. Once all surfaces were installed, each was tested with a servo tester at least five times over several days ensuring full deflections. Later, all servos were tested during full power ground throttle up

testing. This test would only marginally impact aileron loading; however, significant aerodynamic loads are imposed on all other surfaces.

7.3 Manufacturing Testing

Manufacturing testing was performed to ensure that various components could be built and that manufacturing methods were feasible before they were implemented on the aircraft. Manufacturing testing was also performed to verify the integrity of components once they had been incorporated into the aircraft. This two pronged approach was intended to filter serious manufacturing issues early and to confirm that component manufacturing met quality standards as a whole. These tests were designed over the course of the manufacturing process as new components and issues arose.

7.3.1 Balsa Wrap Test

The team needed to test whether the main wing leading edge could be covered with balsa, and to test several wrapping methods. To accomplish this, small sections of the wing were laser cut and assembled. Balsa sheets were soaked in water until adequately flexible and were then wrapped around the leading edge of the wing segment. The sheet was bent to the shape of the airfoil, first without applying leading edge pressure, then while applying leading edge pressure. The third method tested was pre-bending the sheet on a wider circular rod before applying pressure to the leading edge and then bending the sheet to the shape of the airfoil, to make bending easier. The team tested methods to hold sheets in place while drying, including holding the sheet in place by hand, using rubber bands to secure the sheet, pinning the sheet (pictured in Fig. 8.2), and using the heat gun to decrease drying time.

7.3.2 Trailing Edge UltraCote Test

The trailing edges of the main wing, Fowler flaps, and ailerons were not strong enough to withstand the pressure exerted by the UltraCote when heat-shrunk. Small sections of the wing were laser cut, assembled using various reinforcement methods described in Table 7.1, and coated.

Table 7.1: Reinforcement methods for the trailing edge.

Trailing Edge Reinforcement Method	Method
No Reinforcement	Ribs with no connector between trailing edges
Toothpick Reinforcement	Toothpick cross-bracing
Thin Balsa Reinforcement	½ in. balsa sheets sanded to match trailing edge
Medium Balsa Reinforcement	1 in. balsa sheets sanded to match trailing edge
Multigrain Balsa Reinforcement	Medium Balsa Reinforcement and rib bracing in 2 directions

7.3.3 Flutter Test

The team performed a torsion test on the tail to gauge flutter risk. The team could not be sure how rigidly attached the tail would be until both it and the fuselage were built and attached. At that stage, the team was unable to perform any analysis on the tail. As a result, the team measured whether the stiffness of the tail was broadly acceptable or broadly unacceptable based on past experience.

7.3.4 Wingtip Test

The team performed a wing tip test. To perform the test the team lifted the plane with multiple people supporting the wing and then slowly removed support until just the wingtips were supported. One member also placed a hand on the tail to prevent the plane from pitching downwards.

7.3.5 Bad Weather Test

The team planned to perform a bad weather test to ensure that the aircraft could cope with potential wind and rain in Wichita. The team brought the aircraft outside in both light and moderate rain. This tested the water resistance of the aircraft, how water interacted with holes in the UltraCote for the landing gear and servo control arms, how water affected Kapton tape, and how the aircraft performed in inclement weather.

7.4 Data Collector Testing

The Pixhawk, GPS sensor, and pitot tube were first tested indoors to determine data collection procedures. Next, the GPS satellite lock was tested outdoors and accuracy confirmed by walking around with the GPS. The calibration process was tested several times to ensure all values were initialized correctly. Before each test flight, the system will be calibrated with QGroundControl and tested to ensure the estimated states are accurate. After each flight, the data, recorded with an SD card, is uploaded to Flight Review. Filtering with an extended kalman filter is built in. Produced graphs and maps are sanity-checked to ensure the data is recording correctly and is free of errors.

7.5 Wind Tunnel Testing

The wind tunnel plan consists of three phases: model construction, data collection, and data analysis. For initial testing purposes, a single 3-D printed model served as the *Atlas* model. This half-span, 4:15 scale model was mounted in the wind tunnel test region with a variable angle positioning system and three-component pistol grip sting balance. The mount was fitted to position the fuselage inside the boundary layer of the test region. The team calculated lift and drag characteristics for the proposed aircraft from the axial and normal force readings of the pistol grip balance. After accounting for normal and axial forces occurring due to the model weight, sting balance, and mounting device, the team collected data for several AoA. From this data, C_L and C_D were calculated using MATLAB and overlaid against an inviscid analysis in XFLR5.

7.6 Pre-Flight Checklist

For all flights, the processes described Table 7.2 will be performed to ensure operational safety.

Table 7.2: Flight checklist.

Component/System	Action to take
Environment	☐ Record atmospheric conditions, wind speed, and wind direction
Batteries	☐ Check propulsion battery is securely connected, mounted, and fully charged ☐ Check receiver battery is securely connected, mounted, and fully charged
Airframe	☐ Perform general shakedown checking servo linkages, hinges, and load bearing paths
Mission Specific	☐ Confirm passengers are locked into their seats (M2) ☐ Confirm banner is securely stored (M3)
CG Location	☐ Confirm CG location is within acceptable limits marked on the airframe
Landing gear	☐ Confirm landing gear apparatus is fixed to the airframe and the wear on wheels ☐ Confirm that the aircraft tracks straight
Electrical Subsystems	☐ Perform initialization and confirm data collection for Pixhawk flight data recorder ☐ Check control surface deflections for rate, max deflection, and mapping to control sticks
Propulsion	☐ Check motor, prop direction, and ensure screws/nuts are tight and in good condition ☐ Check throttle power, aircraft vibration, and thrust vectoring system
On Site Safety	☐ Ensure clear takeoff and landing paths ☐ Detail expected flight plan to personnel; ensure all personnel are stationed correctly

7.7 Flight Testing Plan

Flight tests were performed with the first iteration design. Minor adjustments were made to the design after each test flight. The final competition vehicle will take into account all results from the test flights. Table 7.3 details the planned flights of the *Atlas*. These flights may be performed by a prototype or the competition vehicle. As of writing, flight tests 1 and 2 have been completed.

Table 7.3: Planned flight tests.

Objective	Flight Test	Notes
Controlled Flight and Landing	1	Cooper-Harper Score, video required for competition
Landing Gear Change Validation	2	Ensure structural changes improved gear performance
In-Flight Fowler Flap Deployment	2	
Stall and Recovery Test	2	
Bad Weather Flight	2	It started to hail (happened to be flight test 2)
M1 Lap Time Performance	2	
Fowler Flap Takeoff	3	Confirm aircraft stability with Fowler flaps during takeoff
M3 Banner Performance	3	Deploy banner in flight
Takeoff Performance Validation	4	Full throttle with ailerons and Fowler flaps deployed
M2 Max Payload Performance	5	Carry all 21 passengers

8. Performance Results

8.1 Propulsion Testing Results

The propulsion system was tested for thrust and throttle setting for a previous iteration of the motor and battery configuration (before the LiPo battery rule changes): a 380 $K_{V, \text{Nominal}}$ motor running an APC 12 in. diameter, 8 in. pitch propeller on a 12S LiPo battery.

The throttle for a single ESC in the propulsion system was slowly ramped up to approximate steady-state static thrust at each throttle fraction (Fig. 8.1). The maximum thrust from this thrust-throttle curve was then used to tune the internal resistance in the propulsion system to match the static thrust. The motor in this system was used in the 2018-19 DBF competition, so the experimental motor constant was already known (at $323.9 K_{V, \text{Empirical}}$) and used to match the internal resistance [9].

Once maximum static thrust was matched in the simulation, the overall thrust-throttle curve was used to scale dynamic thrust over all throttle values. The simulated propulsion system interpolates between steady-state outputs, so the matching simply entailed scaling the output static thrust at each throttle value to the empirical results, then letting the simulation interpolate and output the desired dynamic results. This testing ensured that the simulated propulsion system closely approximated the empirical relations found in the motor test.

An important testing result was that the ESC would lose calibration with each power cycle. However, most ESCs maintain their calibrations. This resulted in a saturated thrust at too little throttle, seen in Fig. 8.1. Recalibrating ESCs for every flight is impractical; the *Atlas* now uses a different model of ESC from those of this test.

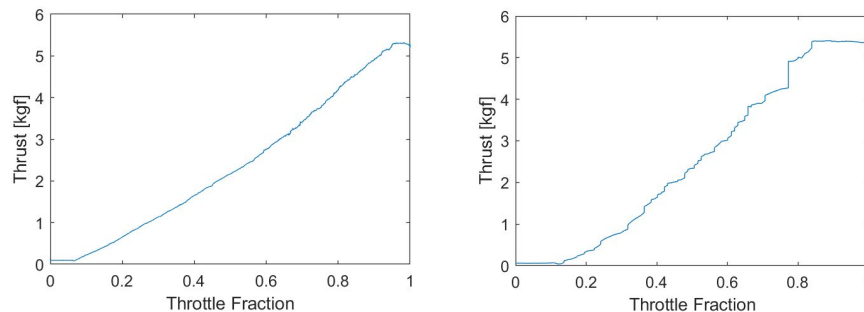


Figure 8.1: Static thrust as a function of throttle for (left) a calibrated ESC, and (right) an uncalibrated ESC.

8.2 System and Subsystem Testing Results

8.2.1 Transmitter Testing Results

The transmitter fail-safe was tested on the flight line. The control surfaces performed as desired, verifying the fail-safe mode. Maximum transmitter range was 3000 ft, by estimation in Google Earth when reported signal strength dropped below 35 dBm (derived from single strength rule of thumb). Initial tests produced anomalous results of a 300 ft maximum range. However, this was caused by poor receiver antenna placement. The receiver antennas were rearranged to correct this issue and a new pre-test check was implemented to catch the problem earlier. Tests were conducted using the FrSky Taranis X9D Plus transmitter with updated OpenTX software, and an FrSky S8R receiver.

8.2.2 Banner Mechanism

The string released correctly and the mechanism operated as intended. The designated servo was sufficient to actuate the mechanism and will be used. However, the small scale of the mechanism

made reloading the string more difficult than anticipated. There is extra space where the mechanism is mounted, so it will be scaled slightly larger in the next iteration.

8.2.3 Fowler Flaps

Early tests showed the hardwood dowel — about which the elbows rotate — twisted, jamming the mechanism. It was replaced with steel, solving the issue. Further tests revealed weak glue joints between the steel and the elbows. Epoxy and increased surface area did not solve this. Instead, JB Weld was required to improve the bond. Nylon spacers and lithium grease were required on the guide rails for smooth flap deployment. Pins tended to slide out of the flaps after several deployment cycles, causing the mechanism to seize. This was corrected with angled balsa sticks, which force the pins back into position. While retracted, Fowler flap droop occurred. This was solved with “chopsticks” acting to pinch the flaps when retracted. Following this correction, the mechanism consistently deployed correctly. **The flaps deployed and retracted successfully in a full throttle test, showing that they were flight ready.**

8.2.4 Control Surfaces

All control surfaces passed initial servo testing results. However, on the third trial, the rudder failed to fully deflect. This was due to an improper connection between the pushrod and the servo horn. All connections were tightened with Loctite coated screws. This was also applied across all linkages. After this correction, all servos passed additional trials and a full throttle engine test.

8.3 Manufacturing Testing Results

8.3.1 Balsa Wrap Test

A summary of the results of the testing of balsa wrapping is shown in Table 8.1

Table 8.1: Results of Balsa Wrap Tests.

Leading Edge Wrapping Method	Result
No Leading Edge Pressure	Sheet split along leading edge
Leading Edge Pressure	Success, with some initial difficulty achieving the desired curvature
Pre-Bending and Leading Edge Pressure	Success, with slight undesired warping from pre-bending
Leading Edge Drying Method	Result
Hand Held	Not viable due to long drying times
Rubber Bands	Success, unable to scale to full wing
Pins (Fig. 8.2)	Generally successful with some tearing, unable to scale to full wing
Heat Gun	Success, must use low setting to prevent injury

This test indicated leading edge pressure as the preferable method. At scale, rubber bands could break the trailing edge and cannot run lengthwise. Basswood stringers proved too strong for the pins to be used on the wing. Thus the heat gun and hand held methods were used to dry the wood before glue was applied. This test successfully showed that wrapping the leading edge in balsa was feasible.



Figure 8.2: Pinned leading edge.

8.3.2 Trailing Edge UltraCote Test

A summary of the results of testing UltraCote on the trailing edges is shown in Table 8.2

Table 8.2: Results of Trailing Edge Test.

Trailing Edge Reinforcement Method	Result
No Reinforcement	Failure of ribs at trailing edges
Toothpick Reinforcement	Deformed trailing edge due to lack of flat reinforcement
Thin Balsa Reinforcement	Severe warping of trailing edge reinforcement
Medium Balsa Reinforcement	Moderate warping of trailing edge reinforcement
Multigrain Balsa Reinforcement	Minimal/no warping of trailing edge

The multigrain balsa reinforcement method best eliminated warping in the trailing edge. Therefore, variations of this method were employed in the main wing, Fowler flaps, and ailerons.

8.3.3 Flutter Test Results

The team determined that the initial rigidity of the tail was insufficient and that the flutter risk was high. The team predicted that adding balsa cross bracing in the tail would increase rigidity to satisfactory levels. After making those changes, the aircraft met rigidity expectations, and flutter risk was reduced.

8.3.4 Wingtip Test Results

The team successfully completed a partially loaded aircraft wingtip test. The test was performed with battery weight; however, one motor was missing and no passengers were loaded. The test showed additional wing reinforcements were needed. In the next iteration, the tip chord will increase, allowing the CF rod to span the entire wing and plywood supports will be added.

8.3.5 Bad Weather Test

This test confirmed UltraCote is water resistant. The coating performed well in rain and a wintry mix. Some water entered servo holes — the holes will be shrunk. The Kapton tape remained dry. The plane successfully flew in a wintry mix. As such, the design is adequate for bad weather operations.

8.4 Wind Tunnel Testing Results

Data from the wind tunnel testing was used to create more realistic simulations. XFLR5 generated lift and drag data accounts only for induced drag and ignores fuselage effects and therefore underestimating total lift and drag. The experiment accounted for both parasitic and induced drag. Figure 8.3 shows experimentally determined and theoretical C_L and C_D for an AoA sweep. The wind tunnel is shown in Fig. 8.4. Compared to the theoretical results, lift is higher due to fuselage lift.

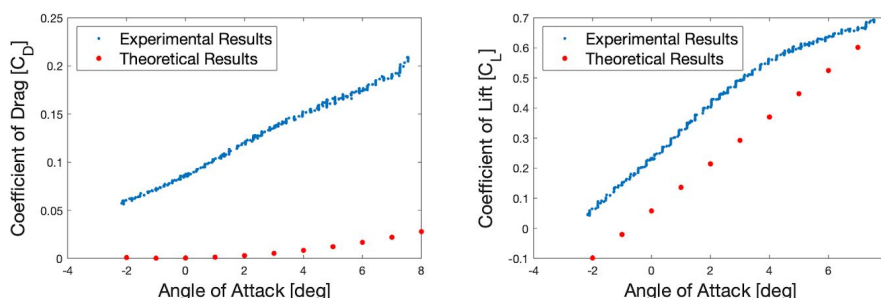


Figure 8.3: Left: C_D for several AoA. Right: C_L as a function of AoA.

The tunnel control speed was 107 mph. Matching Re yielded 12 m/s for the full-scale aircraft. Wing area was used as reference for both C_L and C_D . The C_L is reasonable when compared to the analytical results due to the large fuselage's added lift. C_D is higher than expected even when estimating parasitic drag. This difference is the result of mounting device surface area and surface roughness of the 3D printed model. Results produced were consistent and replicable. Lift values validated XFLR5 results. Drag values informed KM Sim drag models.

To improve KM Sim, wind tunnel lift data replaced XFLR5 data, accounting for the additional lift from the fuselage. A straight section of flight, flown at constant throttle and airspeed, was modeled in KM Sim and drag values were tweaked so that KM Sim results matched flight test results for airspeed, alpha, and throttle setting. With wind tunnel and motor test data, KM Sim accurately matched flight data allowing WUDBF to produce highly accurate performance estimates.



Figure 8.4: Wind tunnel apparatus and setup.

8.5 Flight Testing Results

WUDBF completed two successful flight tests of the *Atlas*. After a receiver failsafe and range check, the preflight checklist (Table 7.2) was completed. During taxi testing, the aircraft experienced catastrophic gear failure. WUDBF switched the landing gear structure to birch plywood and increased the number of cross sections connecting the landing gear platform to the carbon rod.

On 1/17/2020 WUDBF completed the first successful flight of the *Atlas*, shown in Fig. 8.7. The plane tracked well on takeoff but, as velocity increased, the *Atlas* encountered a vibrational mode due to an impulse force. The pilot avoided further complications by quickly increasing throttle, and plans to increase damping within the landing gear structure will solve this problem. The *Atlas* has good handling

characteristics considering its high wing loading. Once trimmed, the plane tracked straight with ample elevator and roll authority. Coordinated turns were required and the pilot felt the vertical tail was undersized. WUDBF will increase its size on the competition vehicle. The pilot noted the rudder control was sensitive. Overall, the aircraft earned a Cooper-Harper score of 3. The aircraft cruised at 20.8 m/s and traveled 6.42 km in 4:15 minutes, reaching a maximum altitude of 137.5 m and a maximum loading of 3.8g. Figure 8.5 details the aircraft's airspeed over time. In response to a landing gear failure on landing, WUDBF increased gear rigidity by shifting the attachment mechanism from a square carbon rod to a plywood sheet between main fuselage bulkheads.

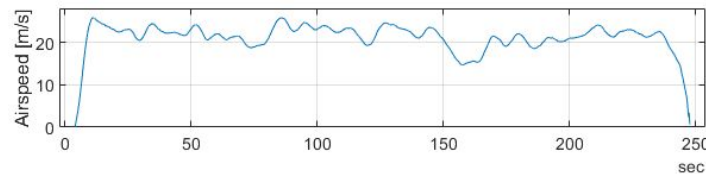


Figure 8.5: Airspeed of the *Atlas* in its maiden flight test.

WUDBF tested the *Atlas* in a second successful flight on 1/31/2020. In this 5:28 flight, the *Atlas* flew 8.23 km, averaging 19.2 m/s with a maximum altitude of 118 m. Figure 8.6 shows the aircraft trajectory. In-flight Fowler flap testing was performed. The Fowler flaps significantly increased lift and should enable a 20 ft take off. The pilot found Fowler flap deployment to induce minimal negative pitching moment, easily compensated for using elevator input. The aircraft experienced controllable benign stall characteristics dropping a wingtip slightly at the very end of the stall. However, due to the downwash built in at the wingtips, the pilot maintained post stall roll control. The *Atlas* has high wing loading, resulting in stall speeds of 13.6 m/s and 9.1 m/s without and with flaps, respectively (See Table 5.7). This requires difficult, high velocity landings. The plane landed with an unequal mass distribution on the main gear, causing the aircraft to roll. This resulted in minor damage to the tail and motor mounts. WUDBF plans to widen the gear base on the competition vehicle for stability and further strengthen the mounting system.

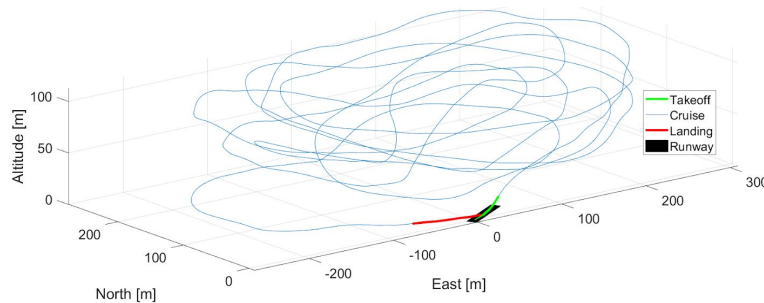


Figure 8.6: Trajectory plot of the *Atlas* using GPS data.

Finally, after the competition vehicle is developed, WUDBF will perform similar system flight checks, ensuring a safe and operable competition vehicle.



Figure 8.7: Takeoff climb of the *Atlas* on a cold winter morning.

9. Bibliography

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